





Cellular Respiration

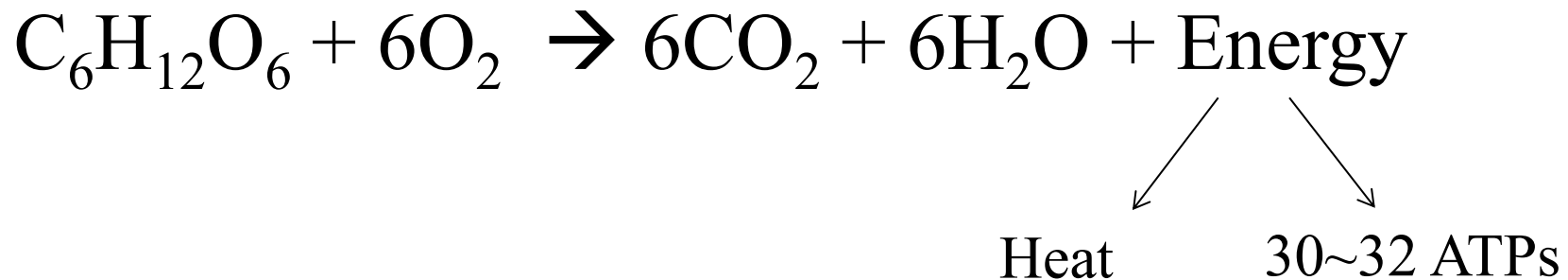


LEHNINGER
PRINCIPLES OF BIOCHEMISTRY

Chapter 14
**Glycolysis, Gluconeogenesis, and
the Pentose Phosphate Pathway**

Cellular Respiration

glucose $\longrightarrow$ energy



Stage 1: Glycolysis (anaerobic) 2ATPs

Stage 2: Citric acid cycle (aerobic) 2ATPs

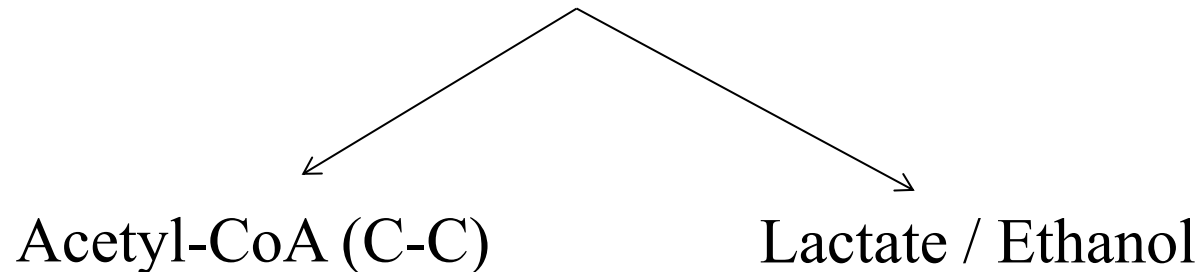
Stage 3: Electron transport chain (aerobic) 26~28 ATPs

Glucose (C-C-C-C-C-C)

Glycolysis

Where ?

Pyruvate (C-C-C)





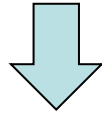
Glycolysis



Preparatory phase

Payoff phase

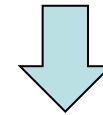
Glucose



G3P, DAP

- 2ATPs

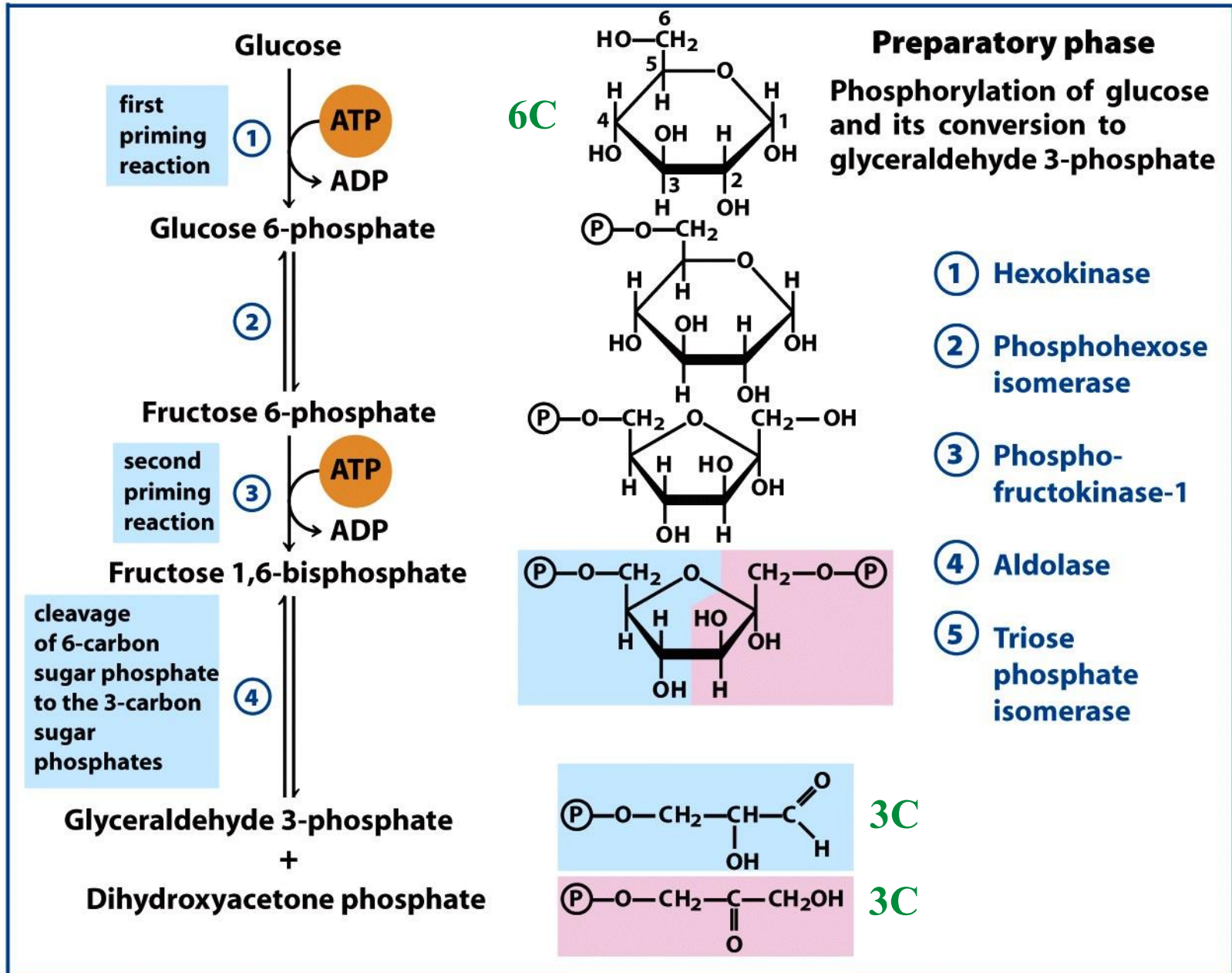
G3P



pyruvate

+ 4ATPs
+ 2NADHs (= ~ 5ATPs)

The two phases of glycolysis



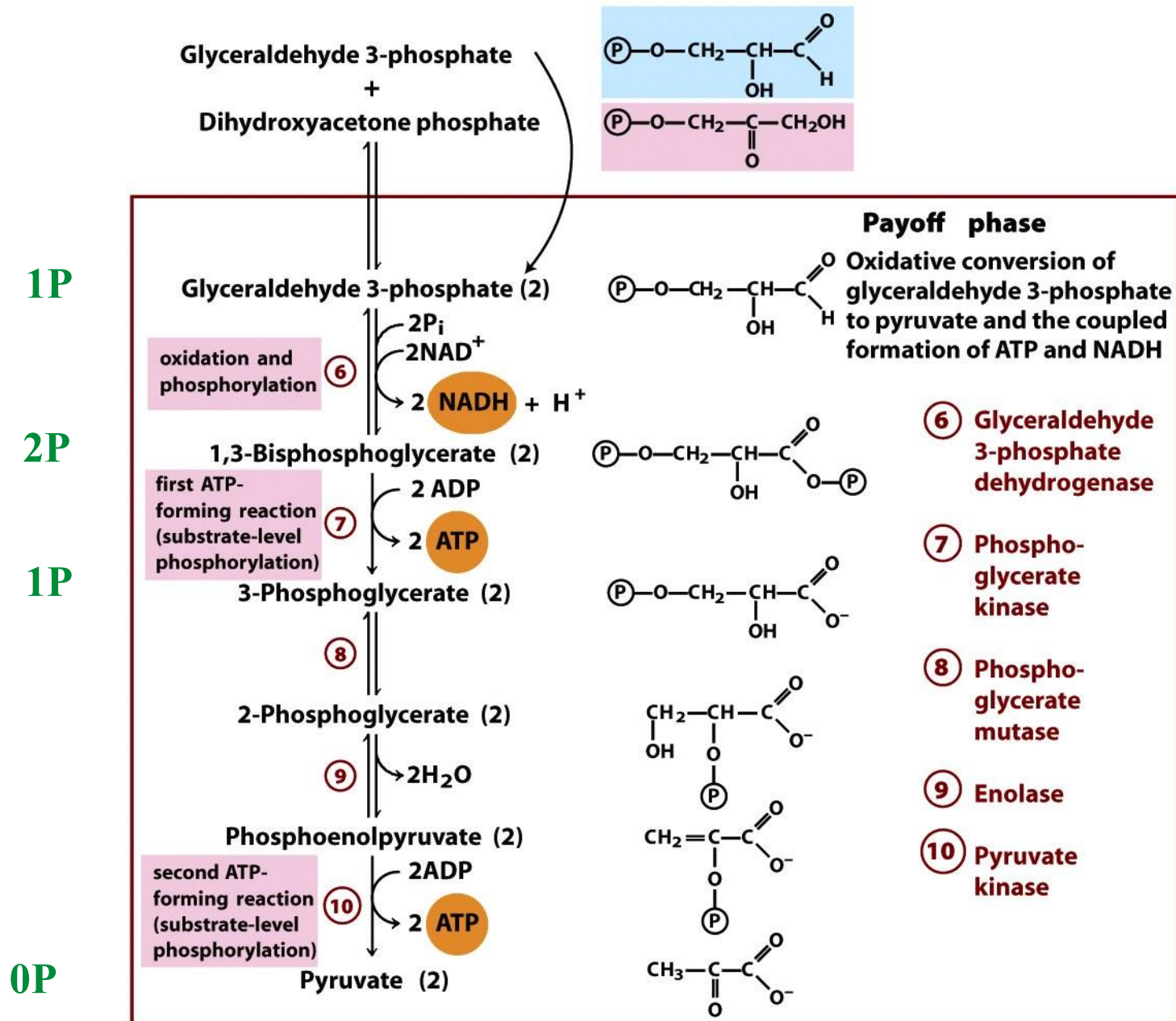
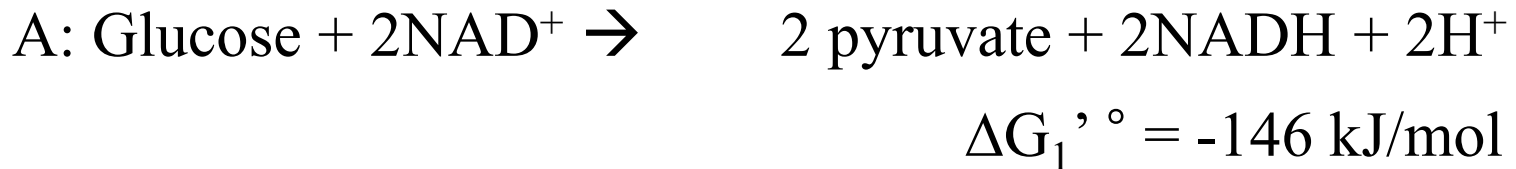


Figure 14-2b

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The overall equation for glycolysis



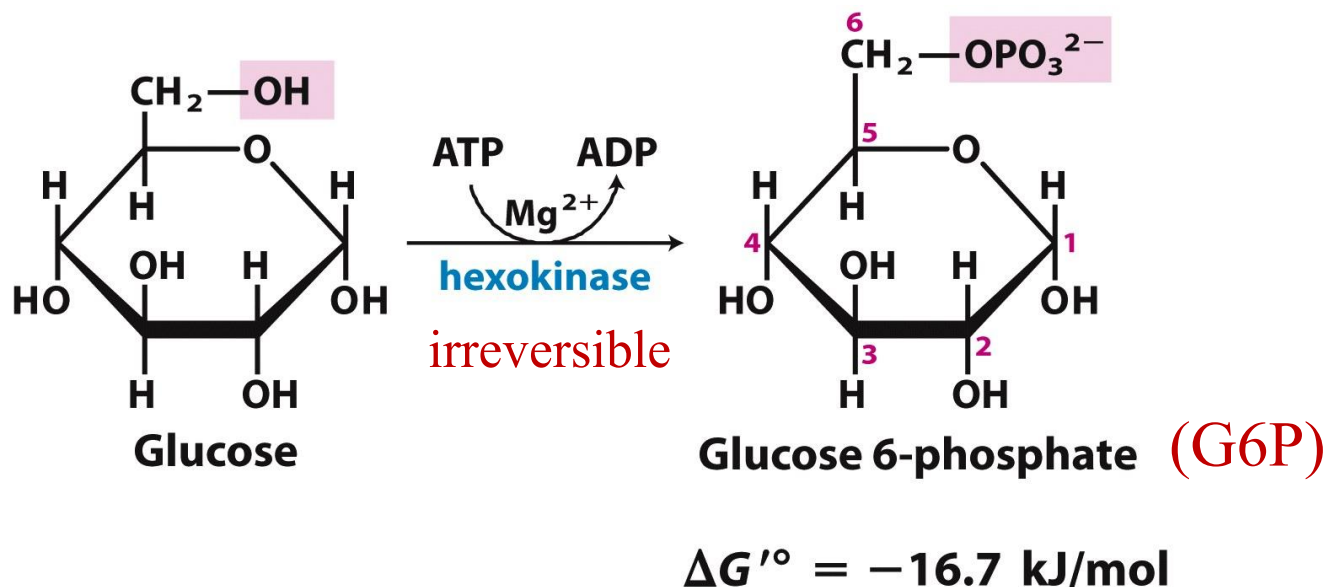
Net yield of ATP = 2ATPs

Energy gain: 2ATPs and 2NADH

The Preparatory phase of Glycolysis

[1] Phosphorylation of glucose (1st priming reaction)

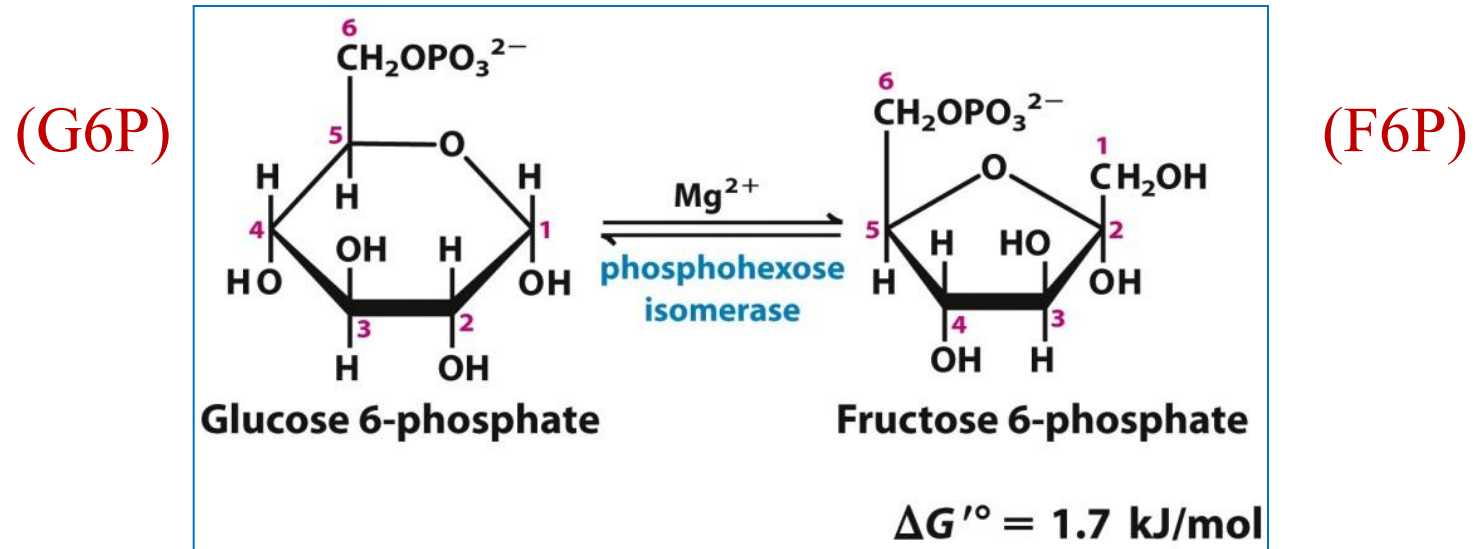
- Mg^{2+} makes the terminal P an easier target for nucleophilic attack
- Hexokinase undergoes an induced fit – brings ATP closer to the glucose
- 4 isozymes (I to IV) of hexokinase is present in human genome



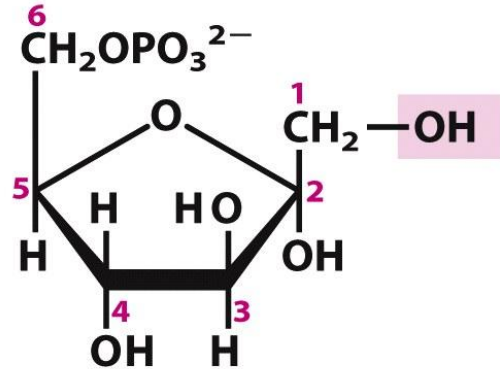
Meaning of phosphorylation for intermediates

1. For conservation of **metabolic energy**
2. Binding of P groups lowers the **activation energy** and increase the **specificity** of the enzymatic reactions
3. The phosphorylated glycolytic intermediates **cannot leave the cell**

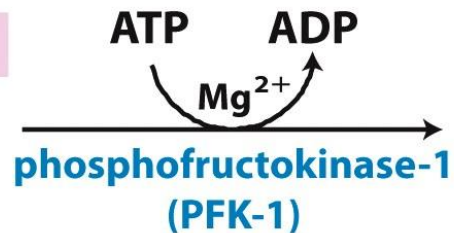
[2] Conversion of glucose-6-P to fructose-6-P



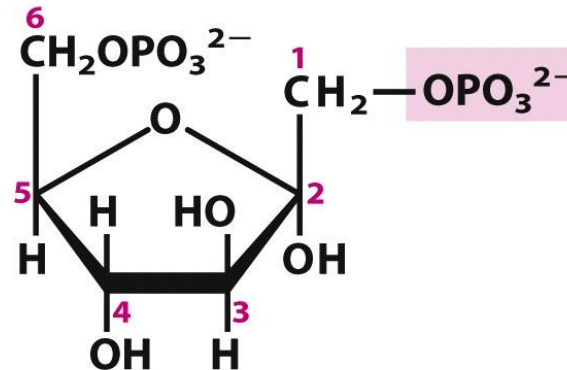
[3] Phosphorylation of Fructose-6-P to Fructose-1,6-bisP (2nd priming reaction)



Fructose 6-phosphate
(F6P)



Irreversible,
1st “committed” step
; FBP is targeted for
only glycolysis

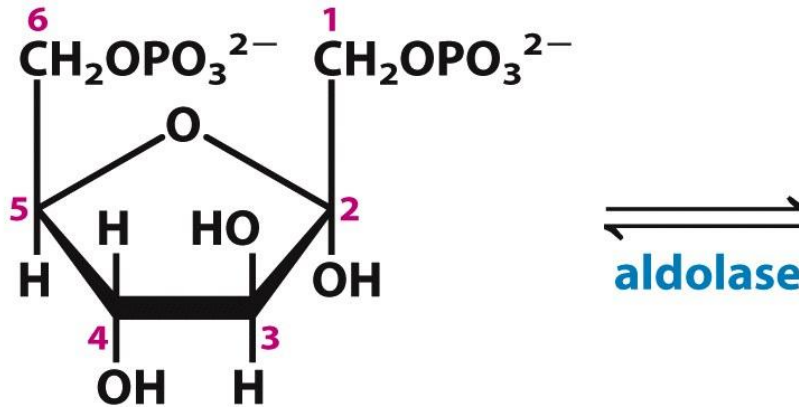


Fructose 1,6-bisphosphate (F16BP)

Q. Difference between
bisphosphate and
diphosphate?

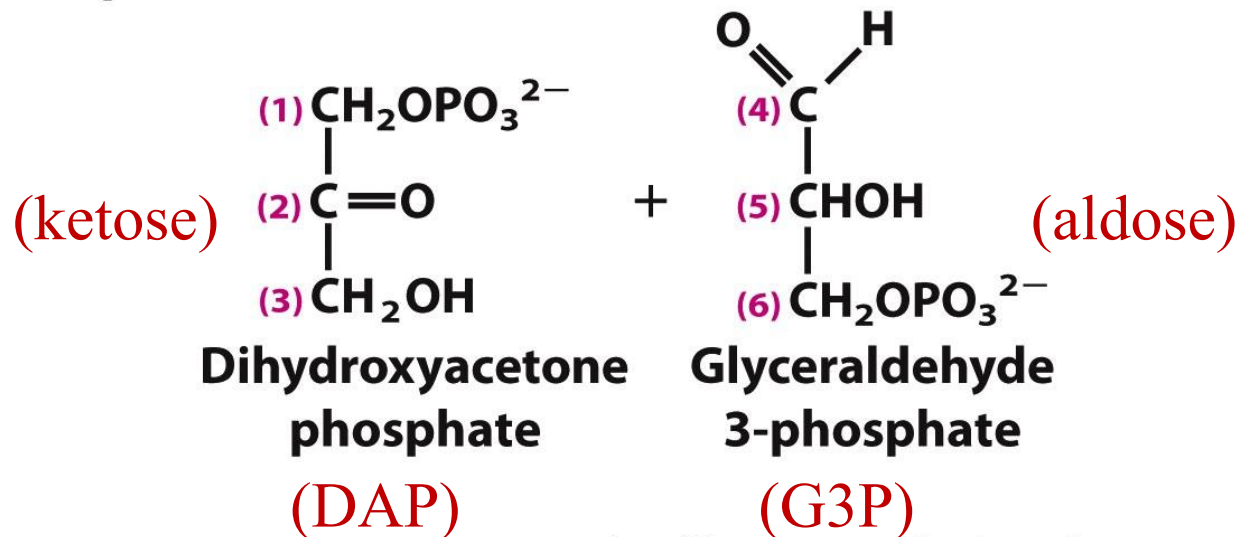
$$\Delta G'^{\circ} = -14.2 \text{ kJ/mol}$$

[4] Cleavage of fructose 1,6-bisphosphate



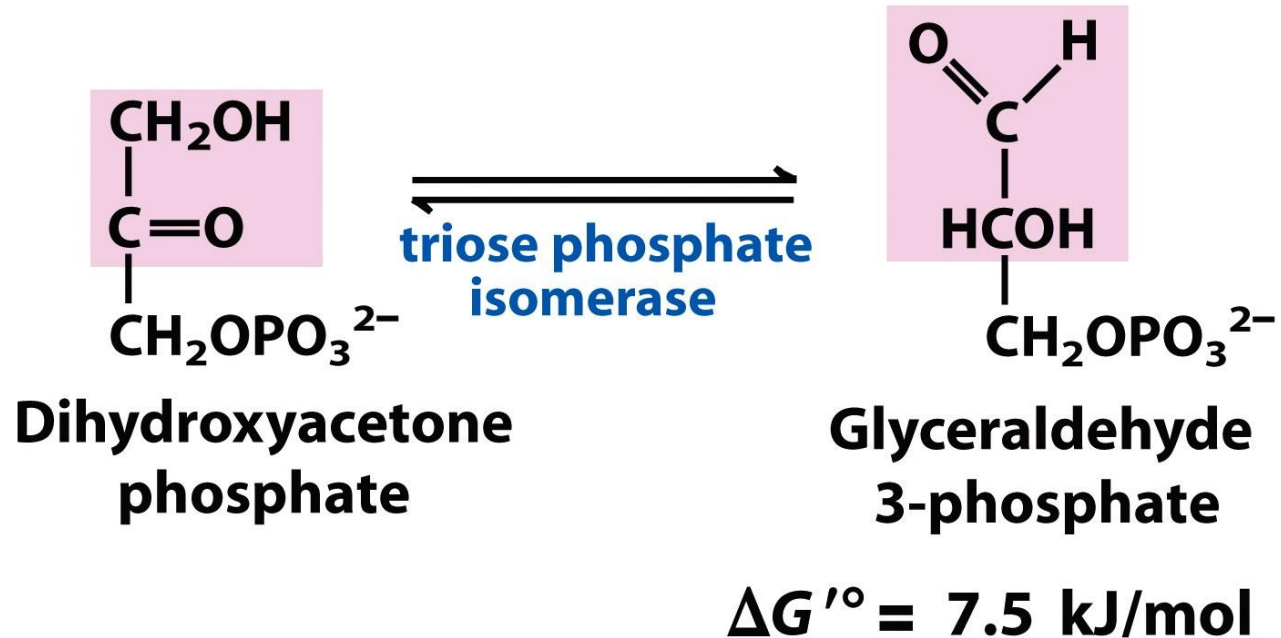
Fructose 1,6-bisphosphate

(F16BP)



$$\Delta G'^{\circ} = 23.8 \text{ kJ/mol}$$

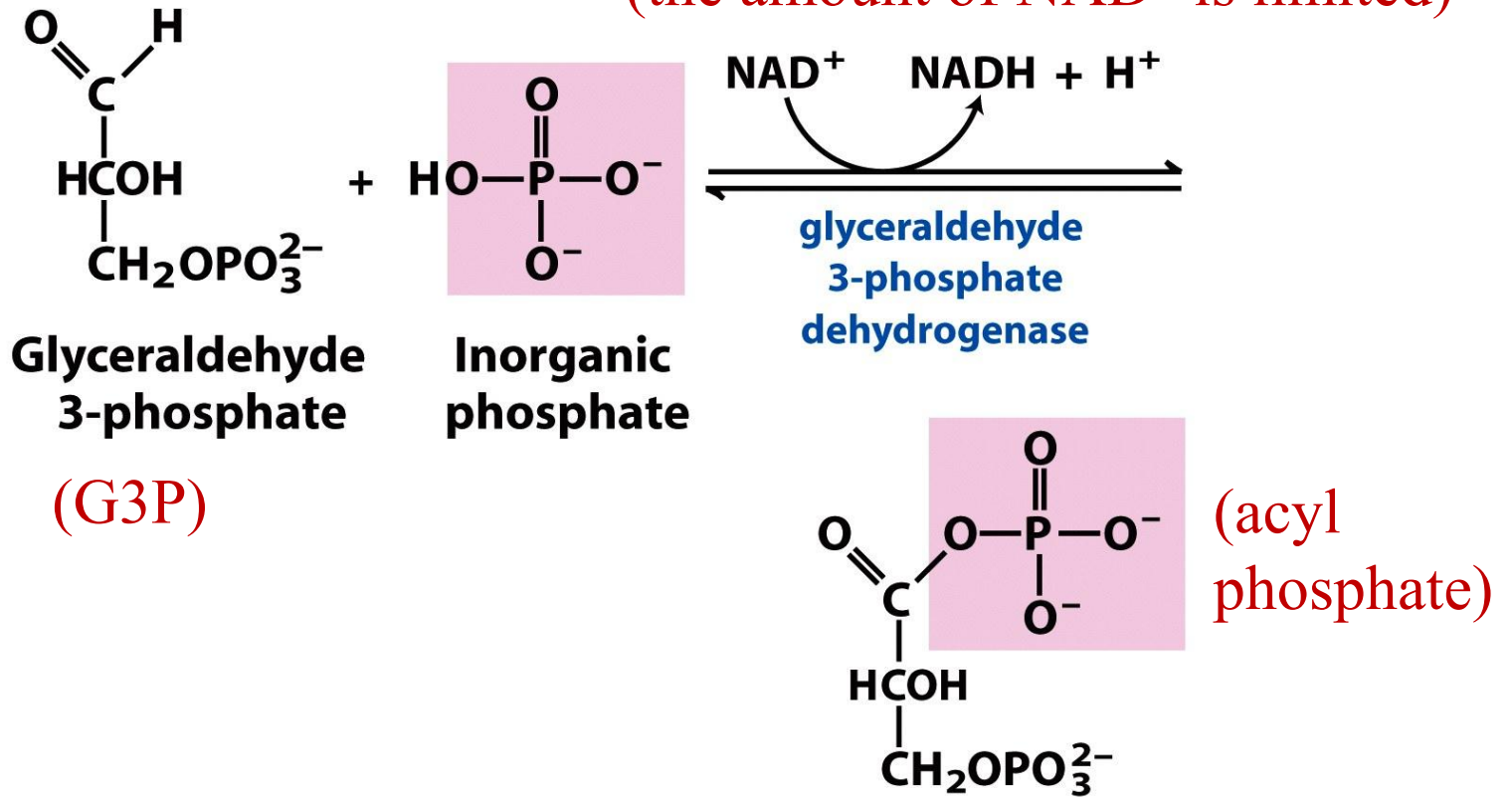
[5] Interconversion of triose phosphates



The Payoff phase of Glycolysis

[6] oxidation of G3P to 1,3-bisphosphoglycerate

(the amount of NAD^+ is limited)

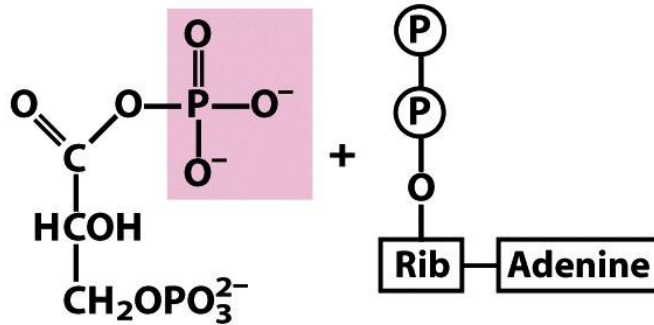


$\Delta G'^{\circ} = 6.3 \text{ kJ/mol}$

1,3-Bisphosphoglycerate (13BPG)

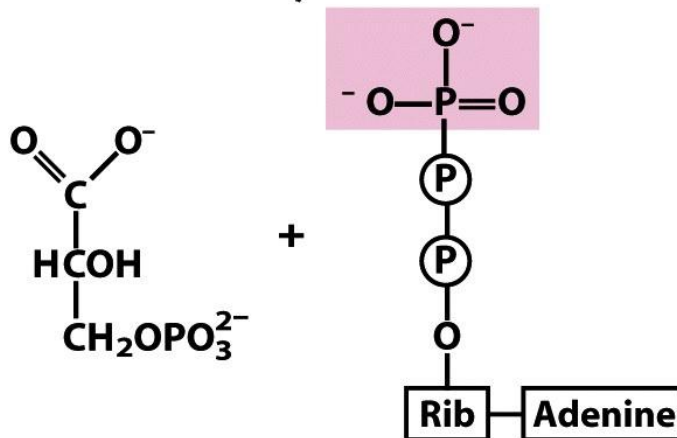
$\Delta G'^{\circ}$ of BPG hydrolysis = -49.3 kJ/mol

[7] Phosphoryl transfer from BPG to ADP



1,3-Bisphosphoglycerate ADP

(BPG)



3-Phosphoglycerate

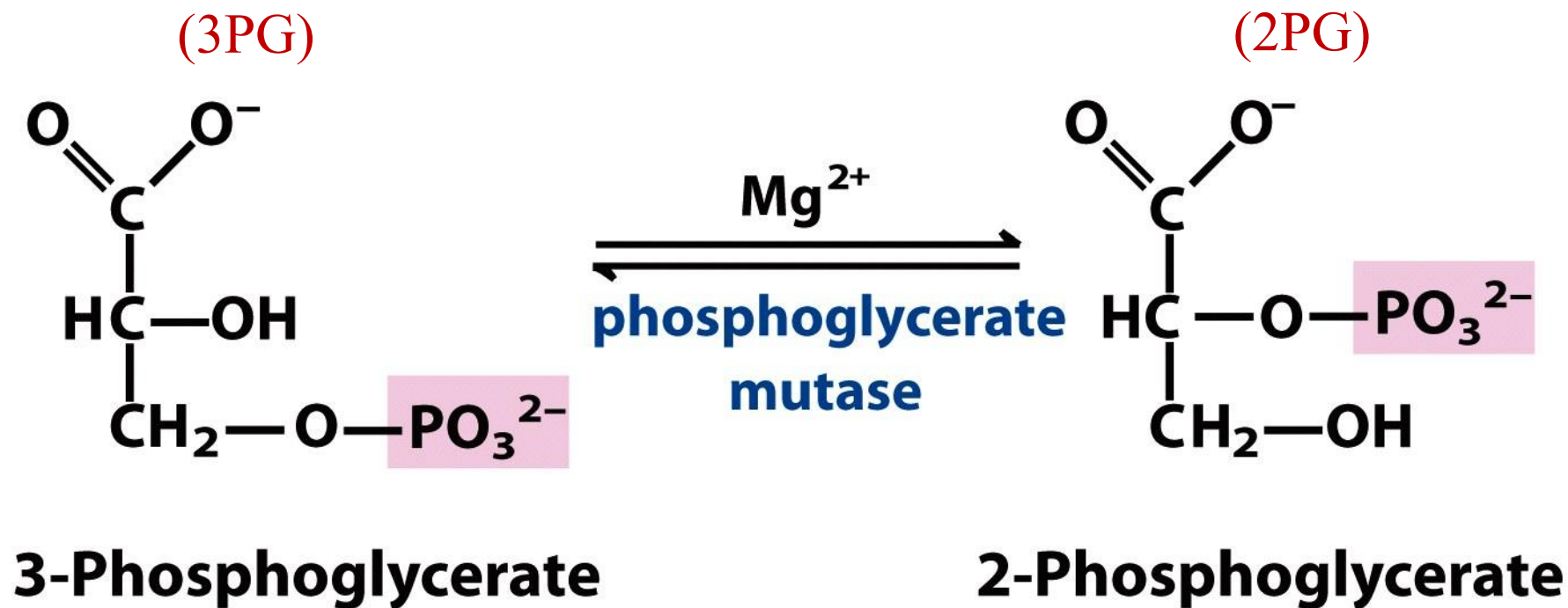
ATP

$\Delta G'^{\circ} = -18.5 \text{ kJ/mol}$

(substrate-level phosphorylation)

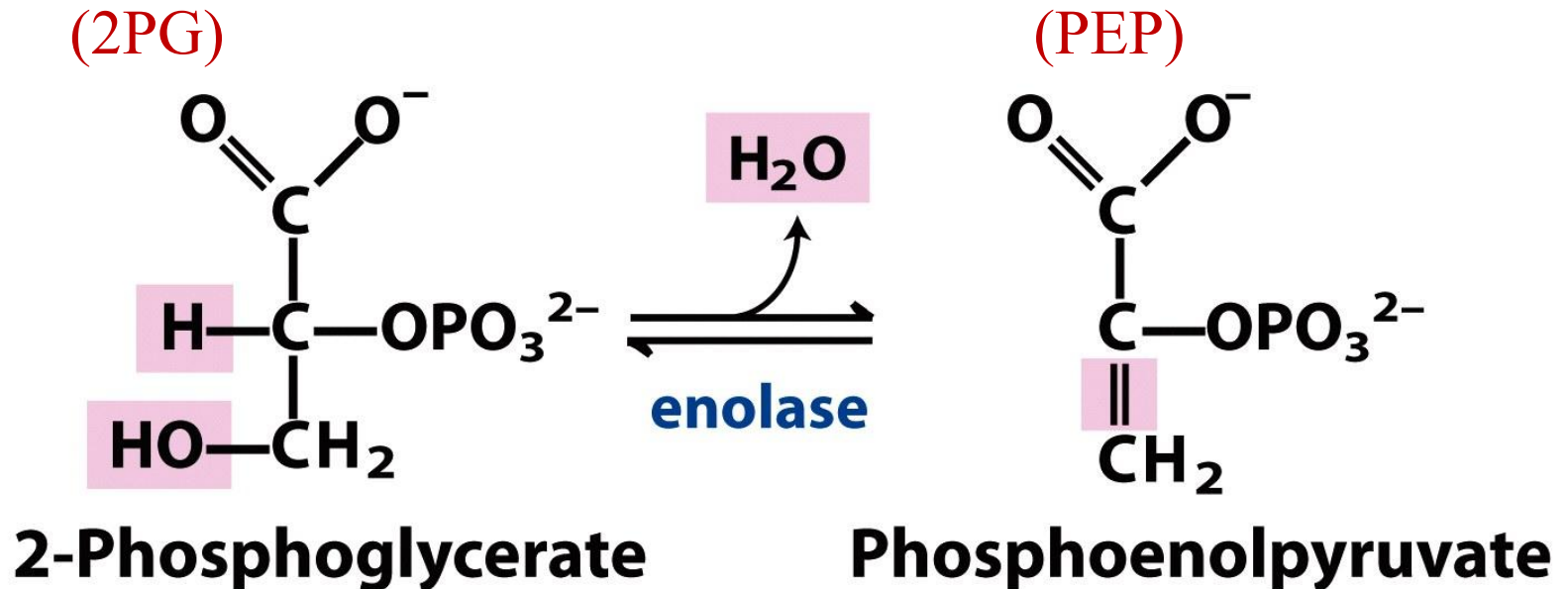
(3PG)

[8] Conversion of 3PG to 2PG



$$\Delta G'^{\circ} = 4.4 \text{ kJ/mol}$$

[9] Dehydration of 2PG to Phosphoenolpyruvate (PEP)

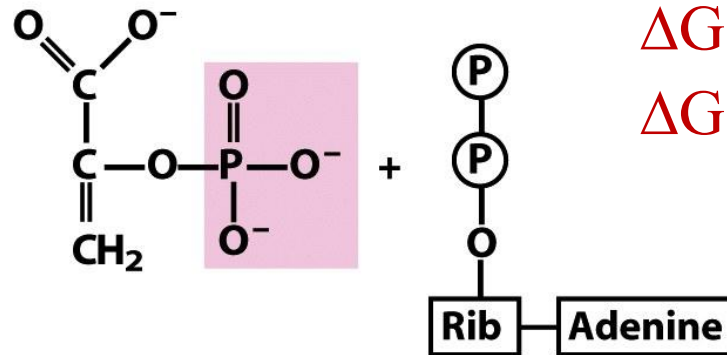


$$\Delta G'^{\circ} = 7.5 \text{ kJ/mol}$$

Q: $\Delta G'^{\circ}$ of 2PG hydrolysis = -17.6 kJ/mol
 $\Delta G'^{\circ}$ of PEP hydrolysis = -61.9 kJ/mol

Enolic intermediate is stabilized by Mg^{2+} → enable to yield PEP efficiently

[10] Transfer of phosphoryl group of PEP to ADP

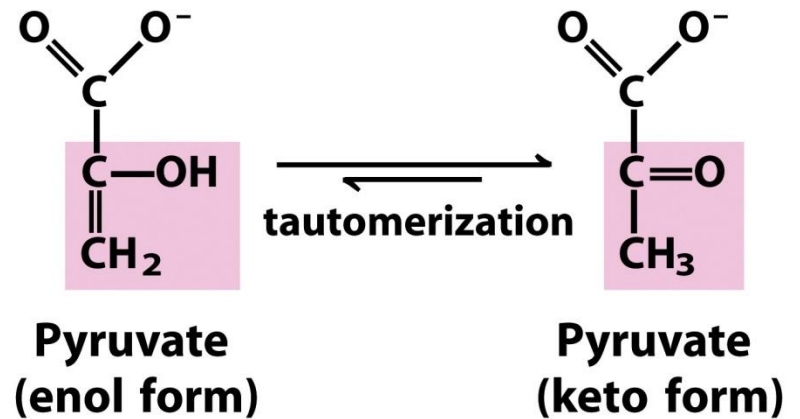
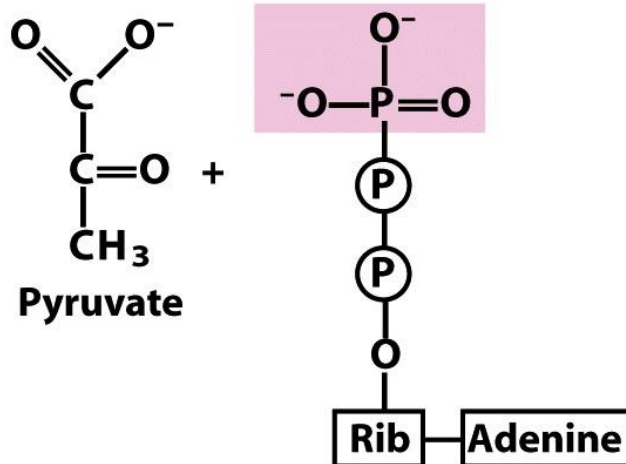


$\Delta G'^{\circ}$ of PEP hydrolysis = -61.9 kJ/mol
 $\Delta G'^{\circ}$ of ATP hydrolysis = -30.5 kJ/mol

Phosphoenolpyruvate

ADP

Mg^{2+}, K^{+} **pyruvate kinase**



Unnumbered 14 p538c
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$\Delta G'^{\circ} = -31.4 \text{ kJ/mol}$ $\xrightarrow{\text{ATP}}$ A large driving force

In anaerobic condition...

“Pasteur effect” : glucose consumption in anaerobic condition is much higher than in aerobic condition

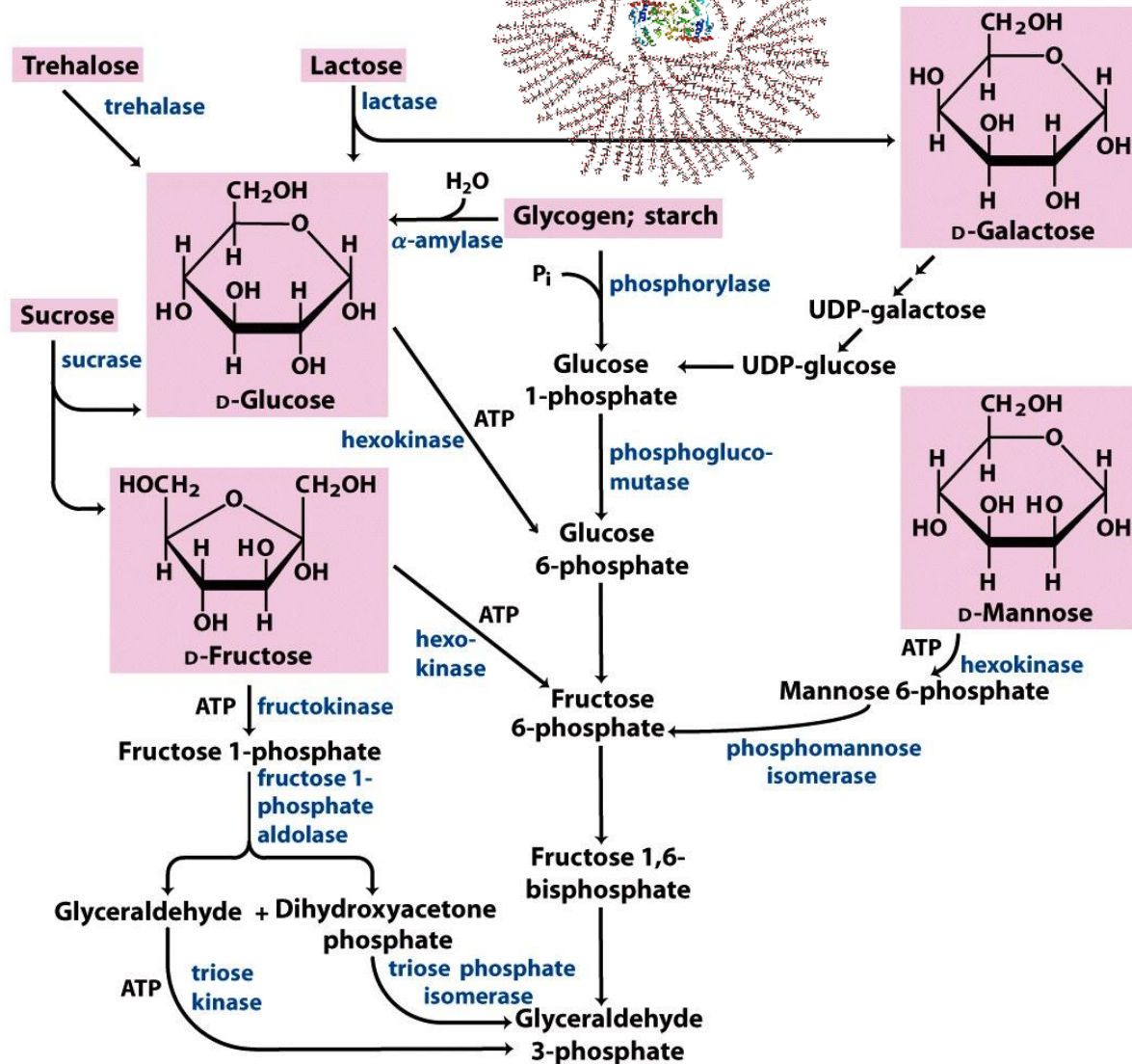
“Warburg effect” : tumors carry out glycolysis at a much higher rate than normal tissue



14.2 Feeder pathways for glycolysis

Entry of the molecules for glycolysis

Entry of dietary glycogen, starch, disaccharides, and hexoses into the preparatory stage of glycolysis

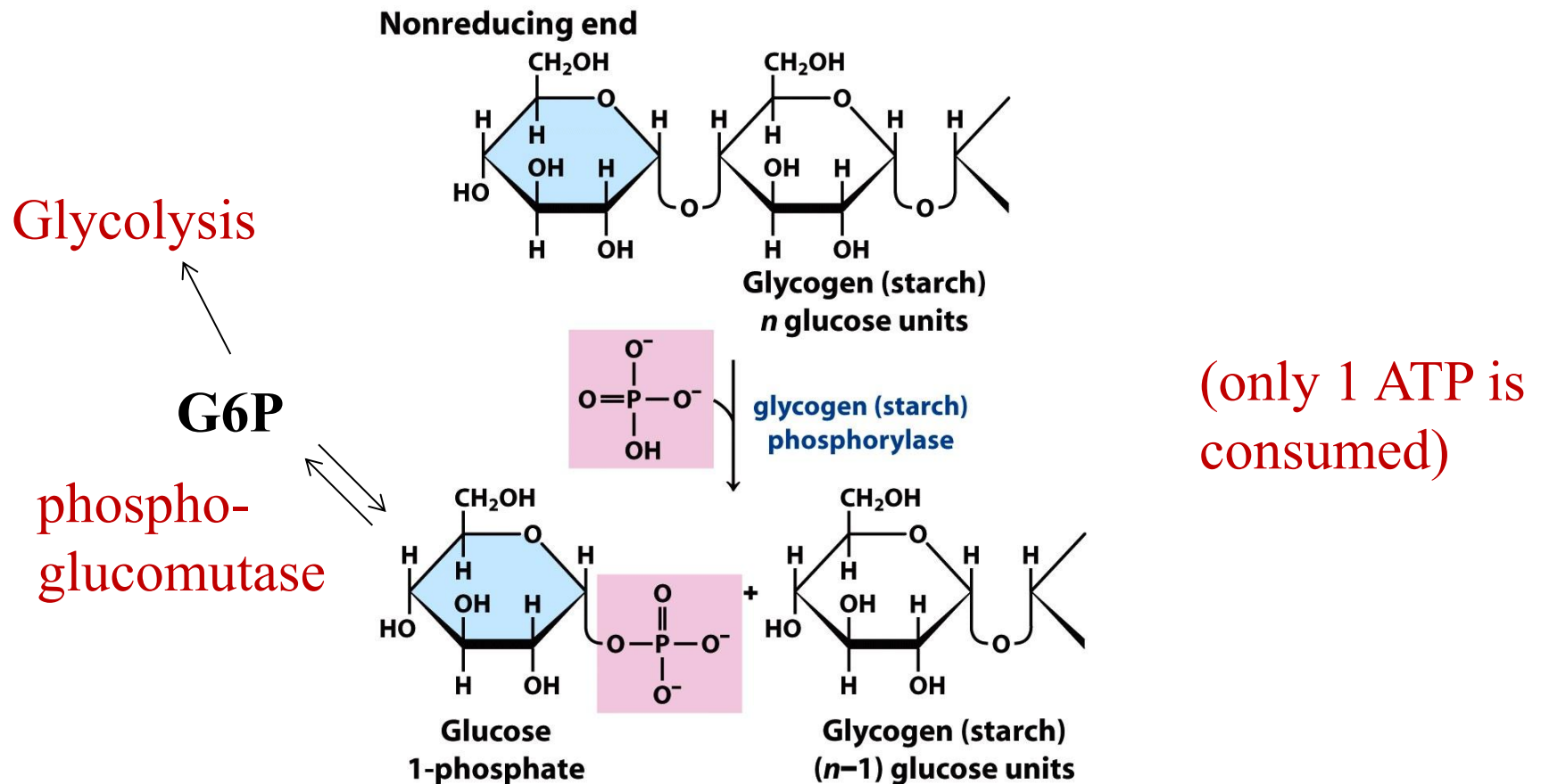


milk



manna

Breakdown of intracellular glycogen by glycogen phosphorylase (**Phosphorolysis**)



Phosphorolysis of intestinal glycogen and starch can not be transported into the cells → Must be dephosphorylated to free sugar

Disaccharides must be hydrolyzed to monosaccharides before entering cells

Dextrin + $n\text{H}_2\text{O} \rightarrow n$ D-glucose (dextrinase)

Maltose + $\text{H}_2\text{O} \rightarrow 2$ D-glucose (maltase)

Lactose + $\text{H}_2\text{O} \rightarrow$ D-galactose + D-glucose (lactase)

Sucrose + $\text{H}_2\text{O} \rightarrow$ D-fructose + D-glucose (sucrase)

Trehalose + $\text{H}_2\text{O} \rightarrow 2$ D-glucose (trehalase)

Conversion of galactose to glucose 1-phosphate

UDP (uridine diphosphate)
: Coenzyme-like carrier of
hexose groups

If this conversion
is defected,

Galactosemia
(high galactose
accumulation in
blood and urine)

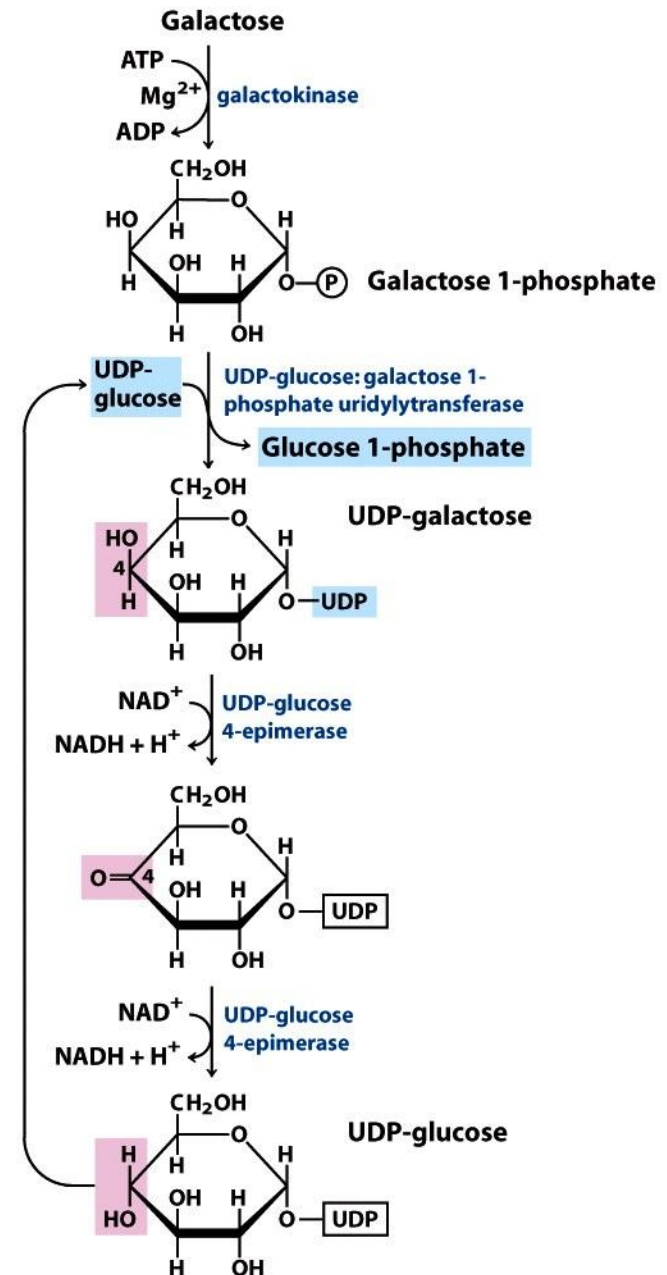


Figure 14-12

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14.3 Fates of Pyruvate under Anaerobic Conditions:

Fermentation



Three possible catabolic fates of the pyruvate formed in glycolysis

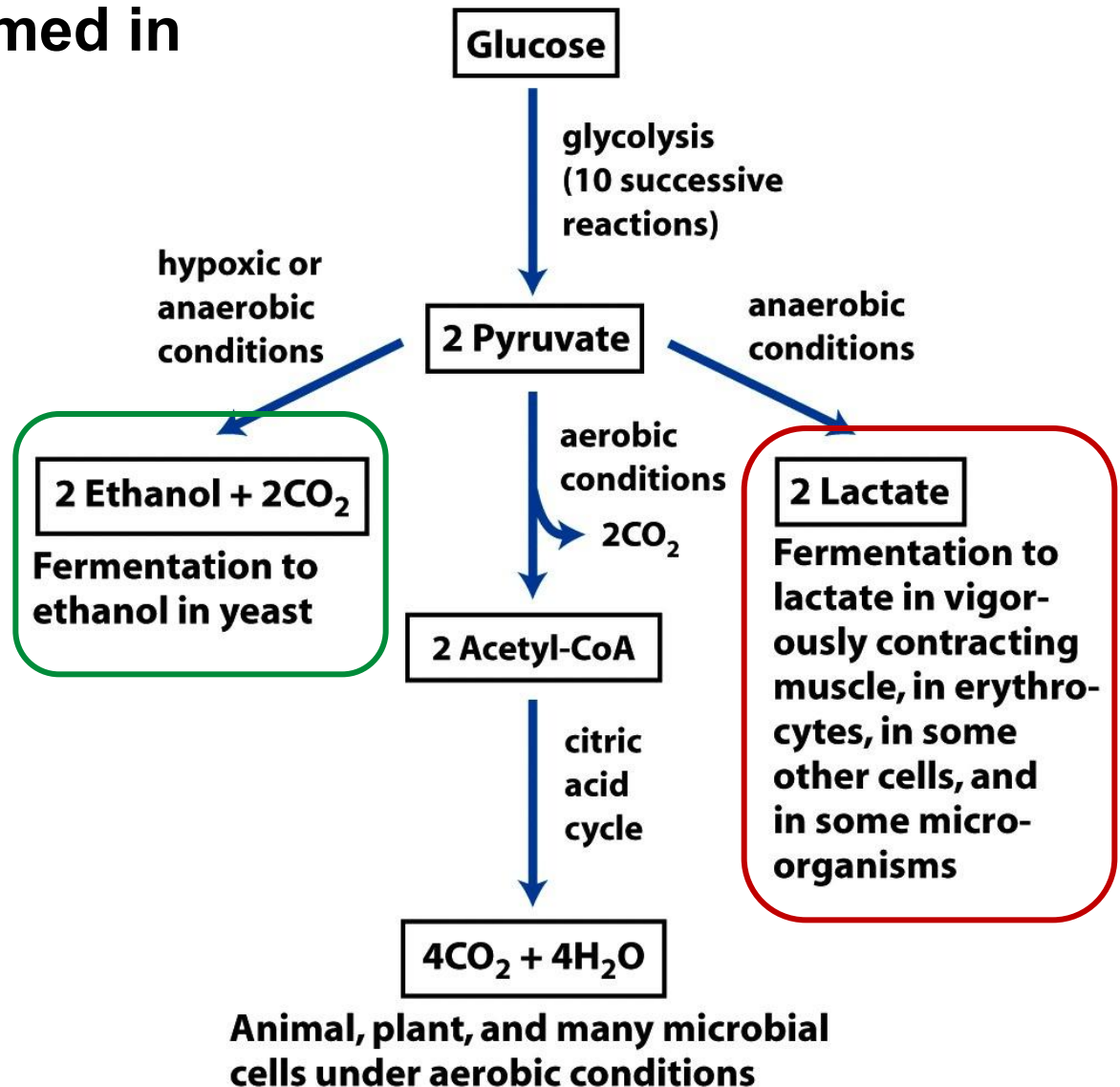


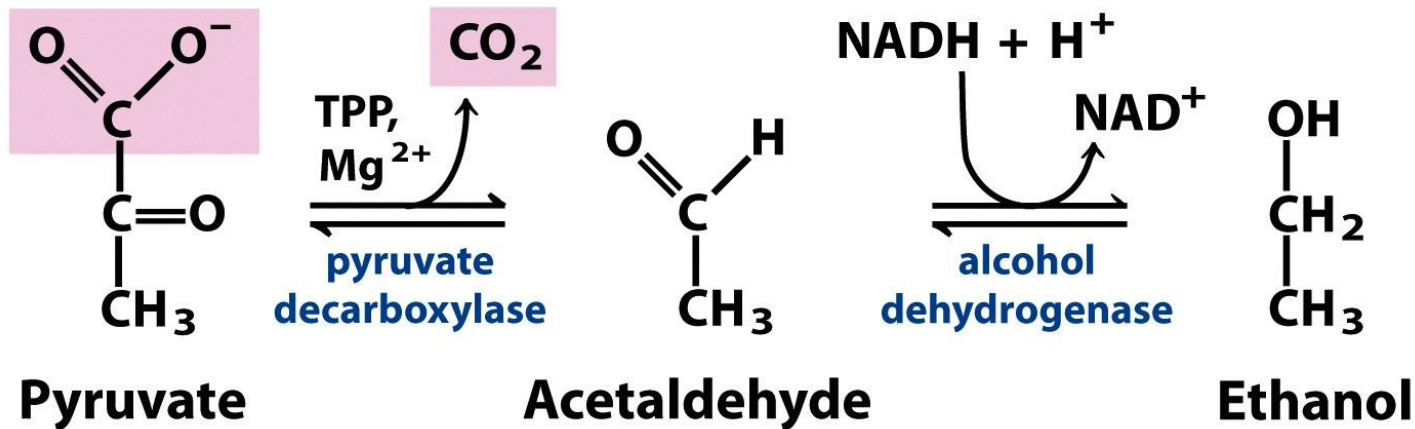
Figure 14-3
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Ethanol Fermentation

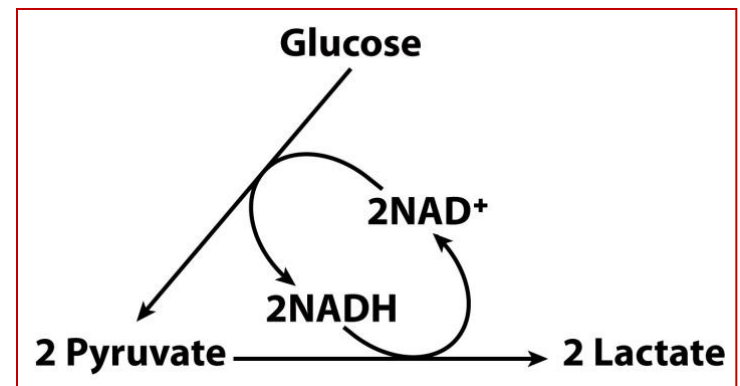
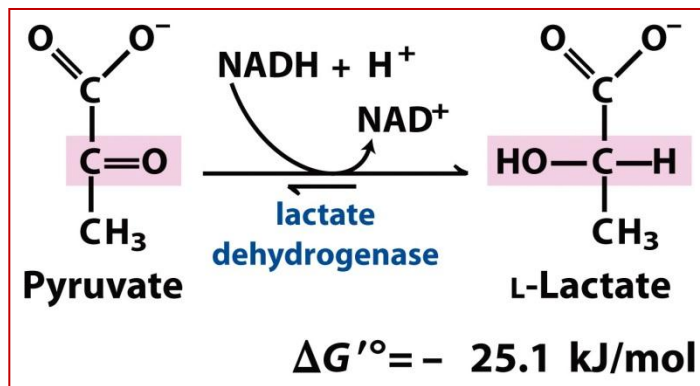


Champagne, baking bread



Fermentation: Extract ATP but do not consume oxygen

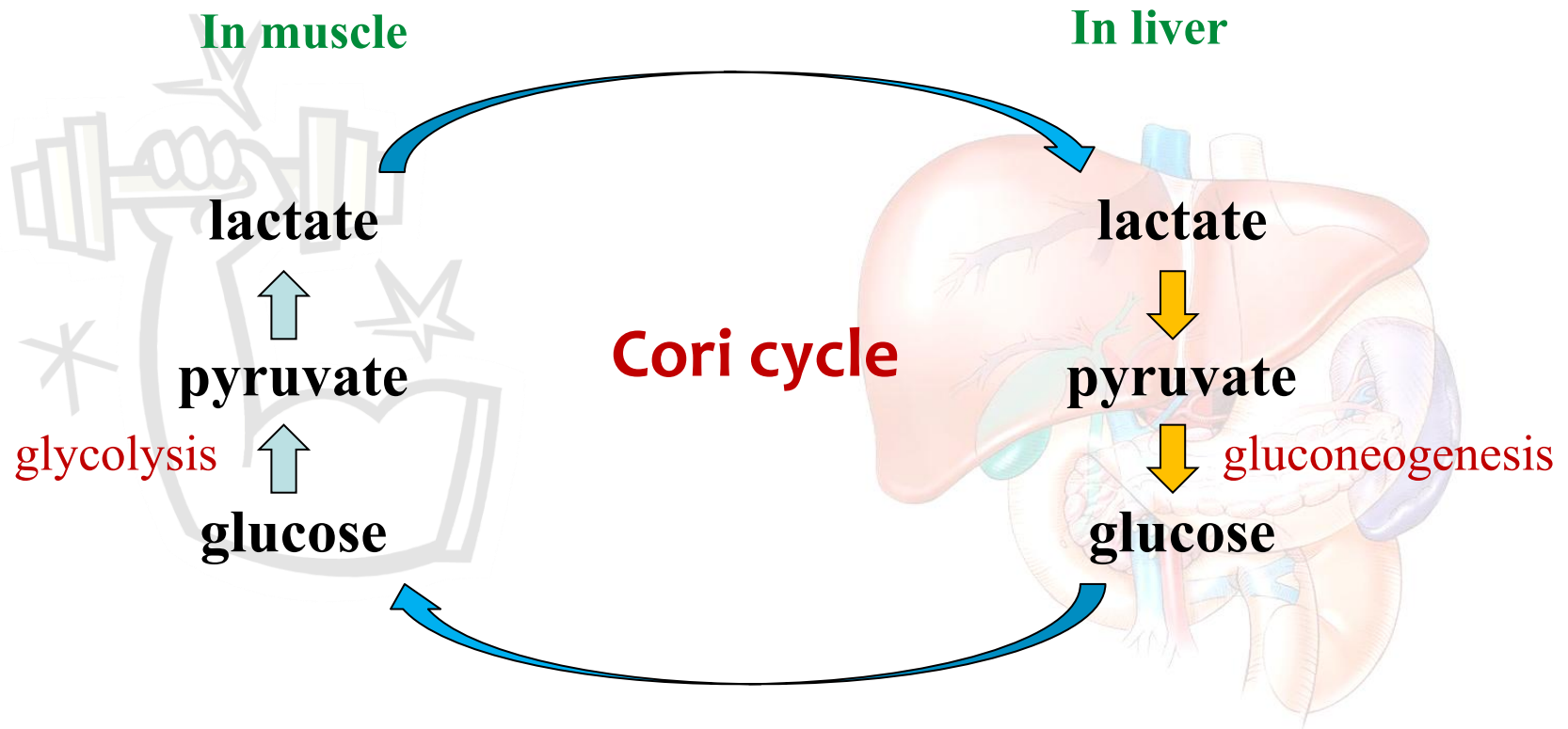
Lactate Fermentation



→ Acidification in muscle and blood

In vigorously contracting muscle, lacking sufficient O_2 , lactate fermentation occurs to provide NAD^+ and to accumulate lactic acid

Lactate can be recycled in liver by Cori cycle



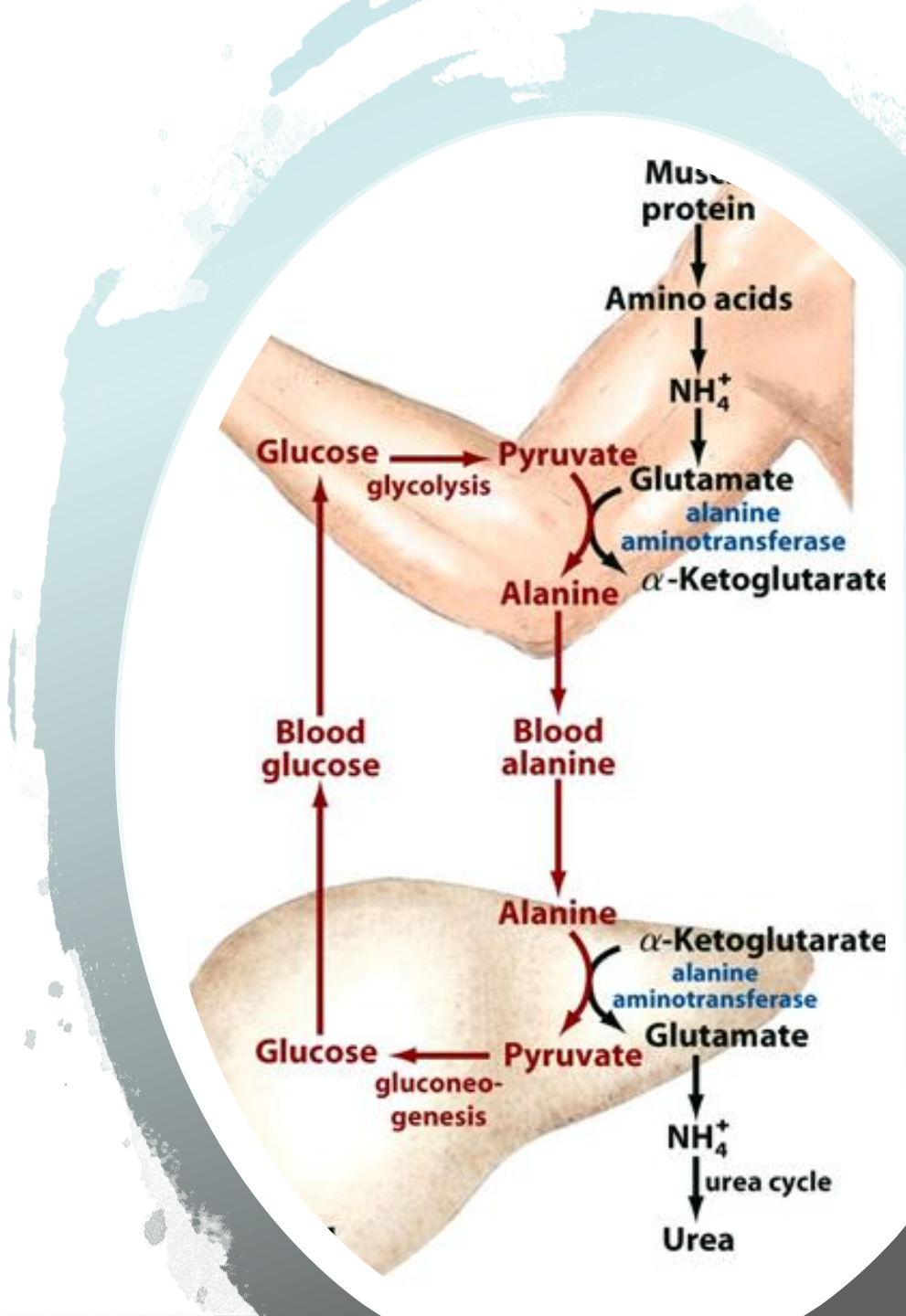
Summary

1. Glycolysis : 1 glucose $\rightarrow$ 2 pyruvate + 2ATP + 2NADH
2. 10 glycolytic enzymes and 10 phosphorylated intermediates
3. Net equation is



4. Glycolysis is tightly regulated to assure an ATP level
5. In type 1 diabetes, defective uptake of glucose cause ketoacidosis

14.4 Gluconeogenesis

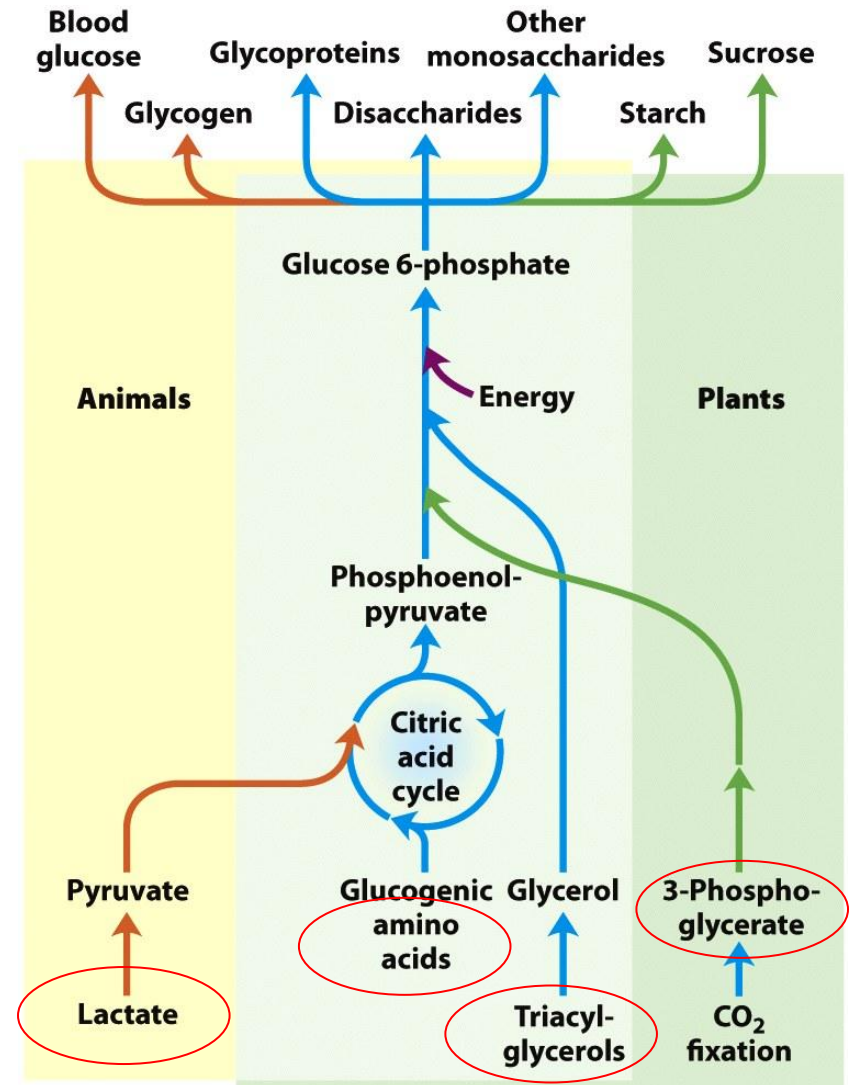


What is Gluconeogenesis?

Carbohydrate synthesis from simple precursors

Glucose consumption in human brain
: 120g /day ~ half of glycogen storage

In case that glucose supply is not sufficient, synthesizing glucose from noncarbohydrate precursors is needed
→ gluconeogenesis
(mainly in liver, and lesser amount in renal cortex and epithelial cells of small intestine)



3 of 10 glycolysis reactions are irreversible with highly negative ΔG values

→ These reactions cannot be used for gluconeogenesis

→ The other 7 reactions can be used for gluconeogenesis

TABLE 14–2 Free-Energy Changes of Glycolytic Reactions in Erythrocytes

Glycolytic reaction step	$\Delta G'^{\circ}$ (kJ/mol)	ΔG (kJ/mol)
1 Glucose + ATP $\longrightarrow$ glucose 6-phosphate + ADP	−16.7	−33.4
2 Glucose 6-phosphate $\rightleftharpoons$ fructose 6-phosphate	1.7	0 to 25
3 Fructose 6-phosphate + ATP $\longrightarrow$ fructose 1,6-bisphosphate + ADP	−14.2	−22.2
4 Fructose 1,6-bisphosphate $\rightleftharpoons$ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate	23.8	−6 to 0
5 Dihydroxyacetone phosphate $\rightleftharpoons$ glyceraldehyde 3-phosphate	7.5	0 to 4
6 Glyceraldehyde 3-phosphate + P_i + NAD^+ $\rightleftharpoons$ 1,3-bisphosphoglycerate + $NADH$ + H^+	6.3	−2 to 2
7 1,3-Bisphosphoglycerate + ADP $\rightleftharpoons$ 3-phosphoglycerate + ATP	−18.8	0 to 2
8 3-Phosphoglycerate $\rightleftharpoons$ 2-phosphoglycerate	4.4	0 to 0.8
9 2-Phosphoglycerate $\rightleftharpoons$ phosphoenolpyruvate + H_2O	7.5	0 to 3.3
10 Phosphoenolpyruvate + ADP $\longrightarrow$ pyruvate + ATP	−31.4	−16.7

Note: $\Delta G'^{\circ}$ is the standard free-energy change, as defined in Chapter 13 (pp. 491–492). ΔG is the free-energy change calculated from the actual concentrations of glycolytic intermediates present under physiological conditions in erythrocytes, at pH 7. The glycolytic reactions bypassed in gluconeogenesis are shown in red. Biochemical equations are not necessarily balanced for H or charge (p. 501).

Opposing pathways of glycolysis and gluconeogenesis in rat liver

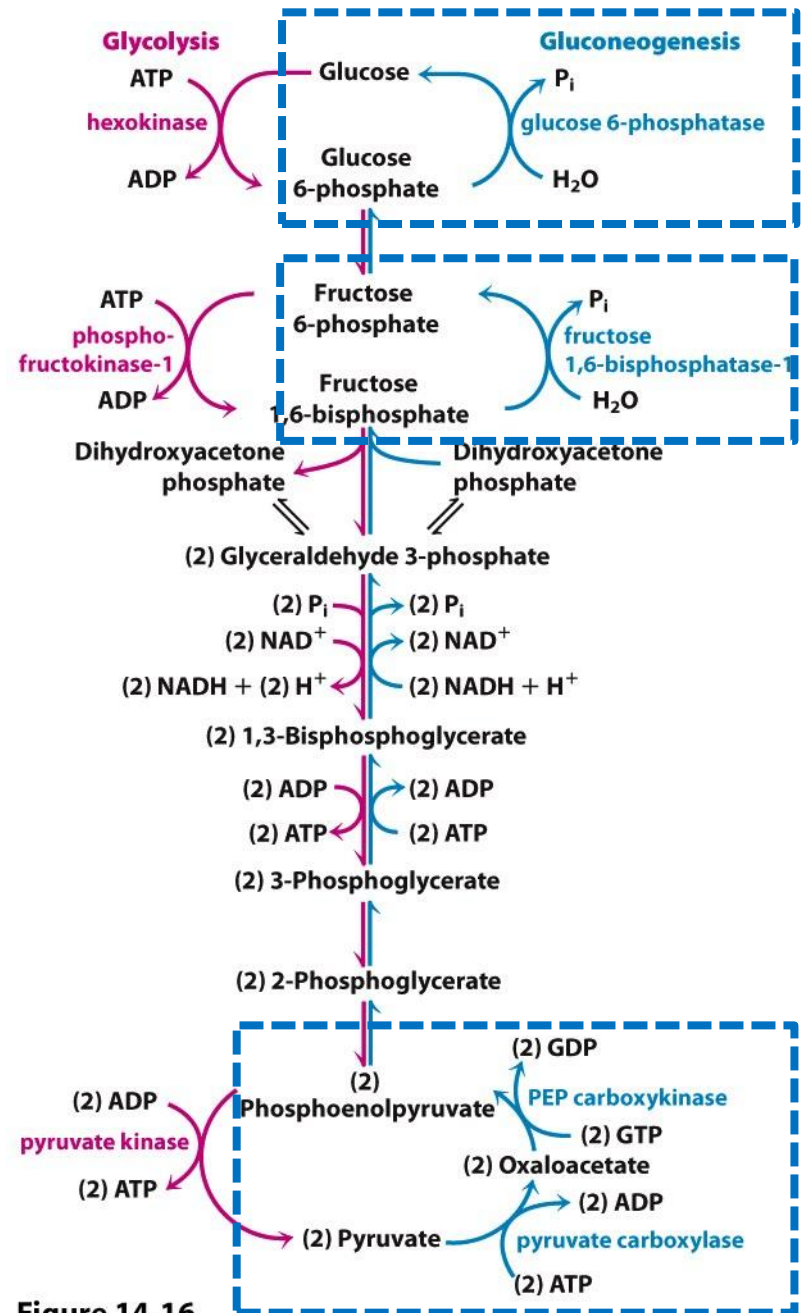


Figure 14-16

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bypass reactions of gluconeogenesis

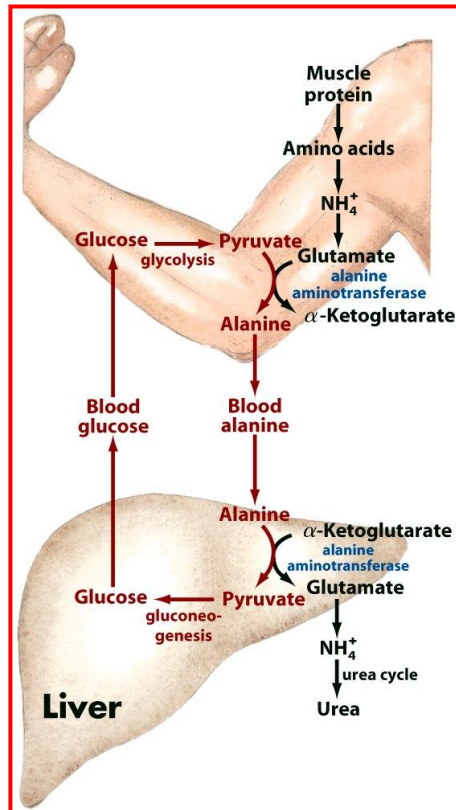
Gluconeogenesis is energetically expensive, but essential

4ATPs + 2GTPs + 2NADHs are needed !

TABLE 14–3	Sequential Reactions in Gluconeogenesis Starting from Pyruvate
Pyruvate + HCO₃[−] + ATP	→ oxaloacetate + ADP + P_i ×2
Oxaloacetate + GTP	⇌ phosphoenolpyruvate + CO₂ + GDP ×2
Phosphoenolpyruvate + H ₂ O	⇌ 2-phosphoglycerate ×2
2-Phosphoglycerate	⇌ 3-phosphoglycerate ×2
3-Phosphoglycerate + ATP	⇌ 1,3-bisphosphoglycerate + ADP ×2
1,3-Bisphosphoglycerate + NADH + H ⁺	⇌ glyceraldehyde 3-phosphate + NAD ⁺ + P _i ×2
Glyceraldehyde 3-phosphate	⇌ dihydroxyacetone phosphate
Glyceraldehyde 3-phosphate + dihydroxyacetone phosphate	⇌ fructose 1,6-bisphosphate
Fructose 1,6-bisphosphate	→ fructose 6-phosphate + P_i
Fructose 6-phosphate	⇌ glucose 6-phosphate
Glucose 6-phosphate + H₂O	→ glucose + P_i
Sum: 2 Pyruvate + 4ATP + 2GTP + 2NADH + 2H⁺ + 4H₂O → glucose + 4ADP + 2GDP + 6P_i + 2NAD⁺	

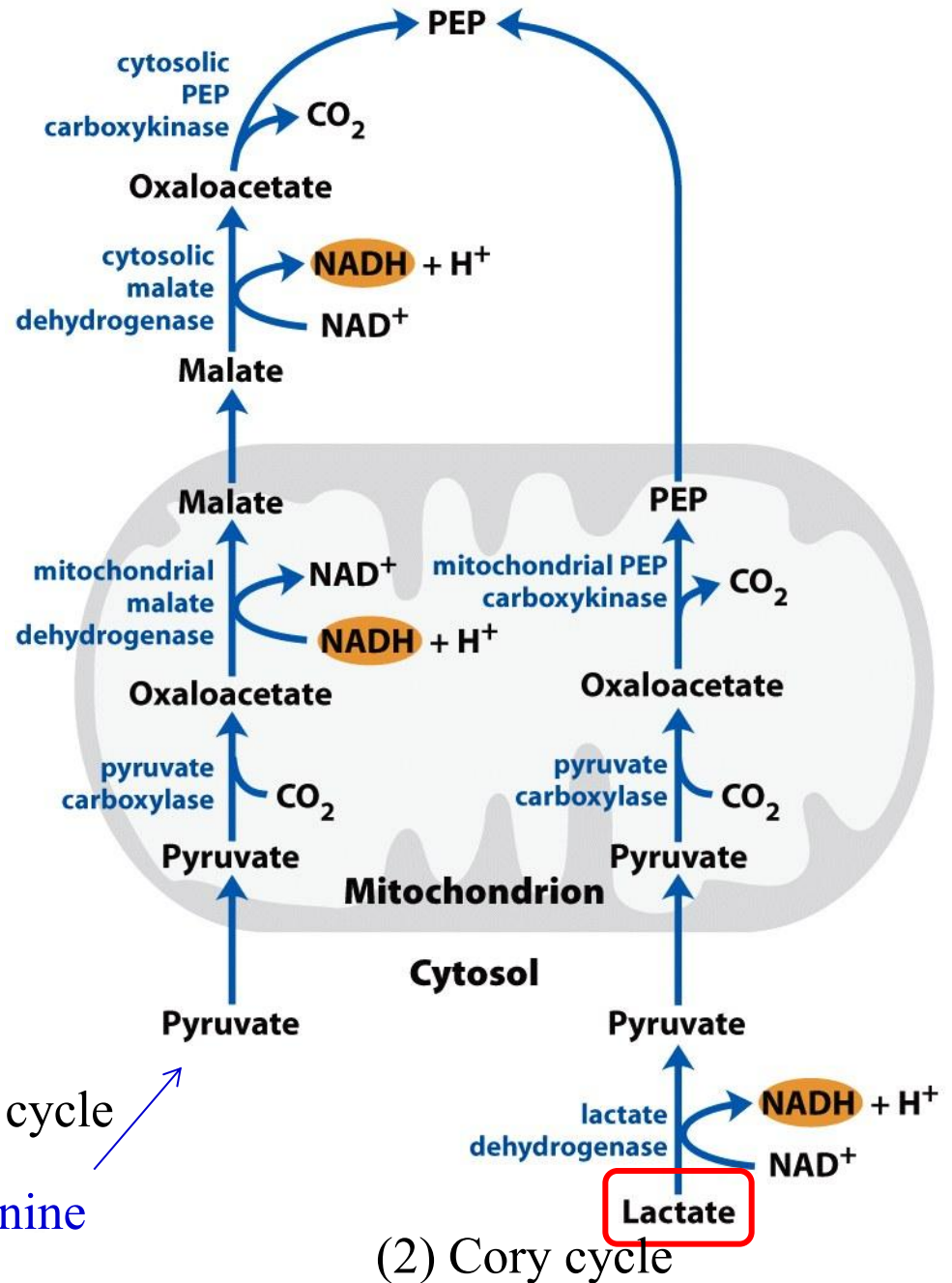
Note: The bypass reactions are in red; all other reactions are reversible steps of glycolysis. The figures at the right indicate that the reaction is to be counted twice, because two three-carbon precursors are required to make a molecule of glucose. The reactions required to replace the cytosolic NADH consumed in the glyceraldehyde 3-phosphate dehydrogenase reaction (the conversion of lactate to pyruvate in the cytosol or the transport of reducing equivalents from mitochondria to the cytosol in the form of malate) are not considered in this summary. Biochemical equations are not necessarily balanced for H and charge (p. 501).

Alternative paths from pyruvate to phosphoenolpyruvate



(1) glucose-alanine cycle

Alanine



(2) Cori cycle

Citric acid cycle intermediates and some amino acids are glucogenic → can be used in gluconeogenesis

TABLE 14–4	Glucogenic Amino Acids, Grouped by Site of Entry
Pyruvate Alanine Cysteine Glycine Serine Threonine Tryptophan*	Succinyl-CoA Isoleucine* Methionine Threonine Valine
α-Ketoglutarate Arginine Glutamate Glutamine Histidine Proline	Fumarate Phenylalanine* Tyrosine*
	Oxaloacetate Asparagine Aspartate

Note: All these amino acids are precursors of blood glucose or liver glycogen, because they can be converted to pyruvate or citric acid cycle intermediates. Of the 20 common amino acids, only leucine and lysine are unable to furnish carbon for net glucose synthesis.

*These amino acids are also ketogenic (see Fig. 18–21).

Fatty acid utilizations of plants and mammals

Plants, yeast, and bacteria

Fatty acids → acetyl-CoA

Acetyl-CoA



glyoxylate cycle

oxaloacetate



gluconeogenesis

Mammals

Triacylglycerols

→ glycerol

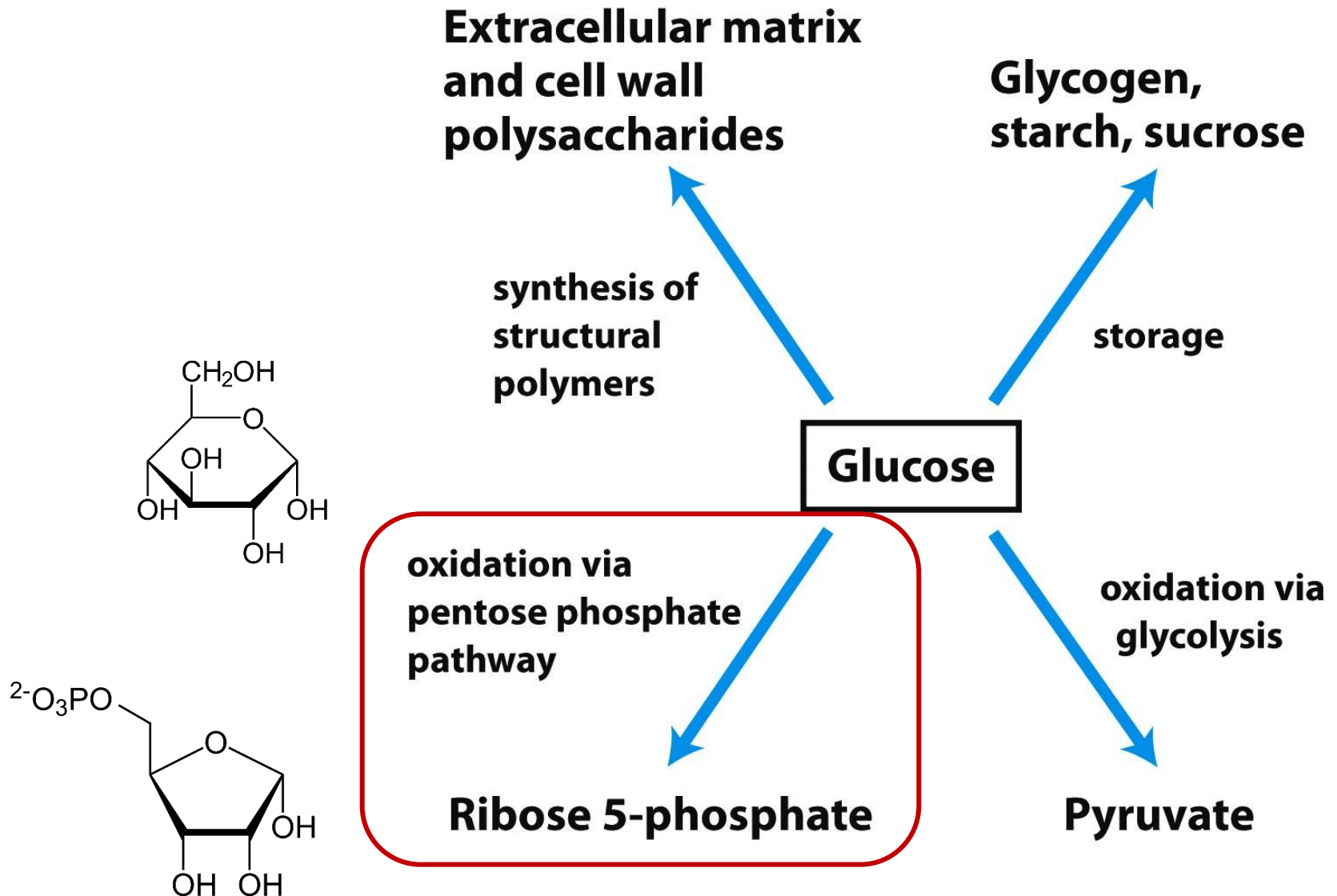
→ Glycerol phosphate

→ Dihydroxyacetone phosphate
(DAP)

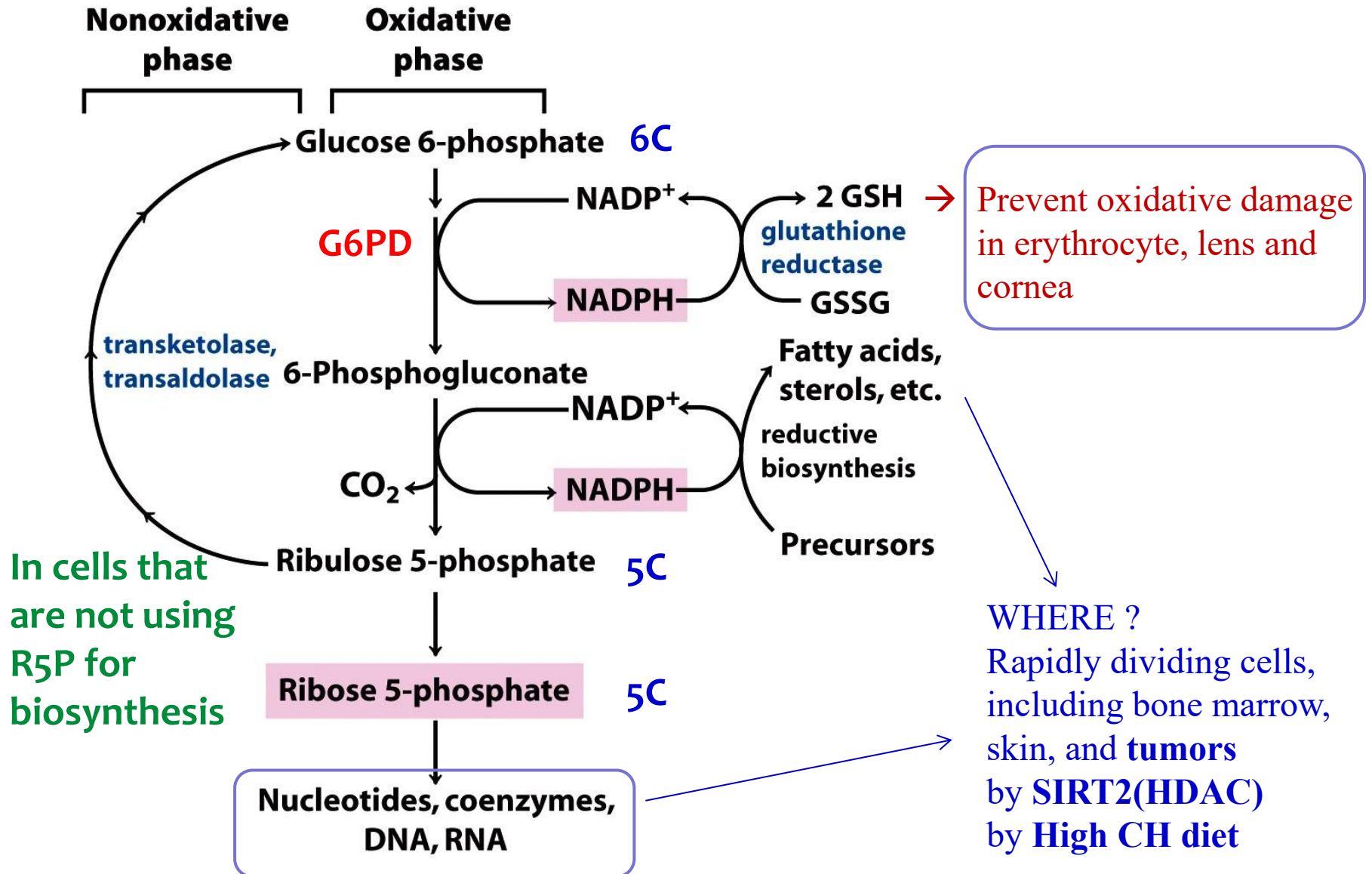
→ gluconeogenesis

14.5 Pentose Phosphate Pathway

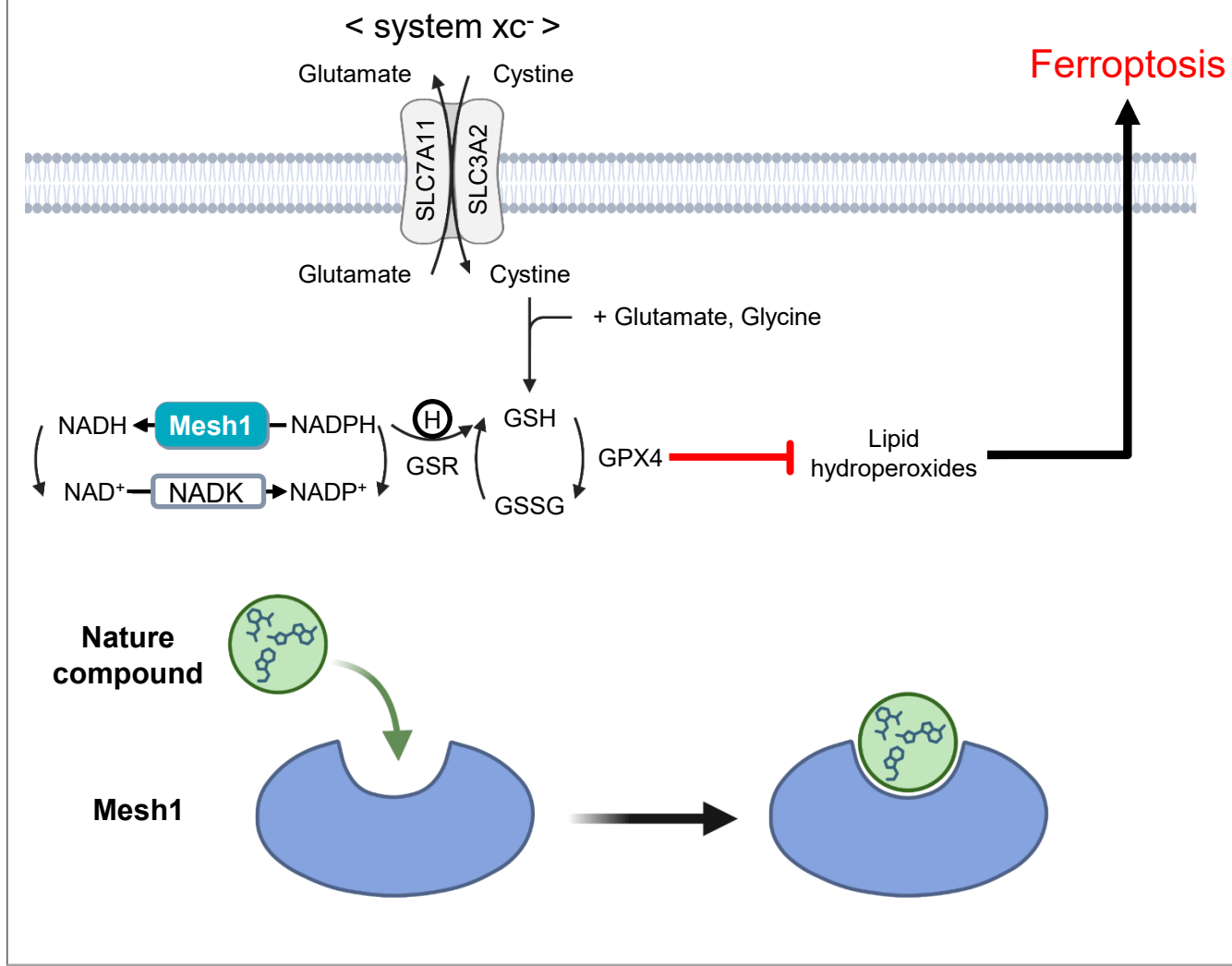
Major pathways of glucose utilization



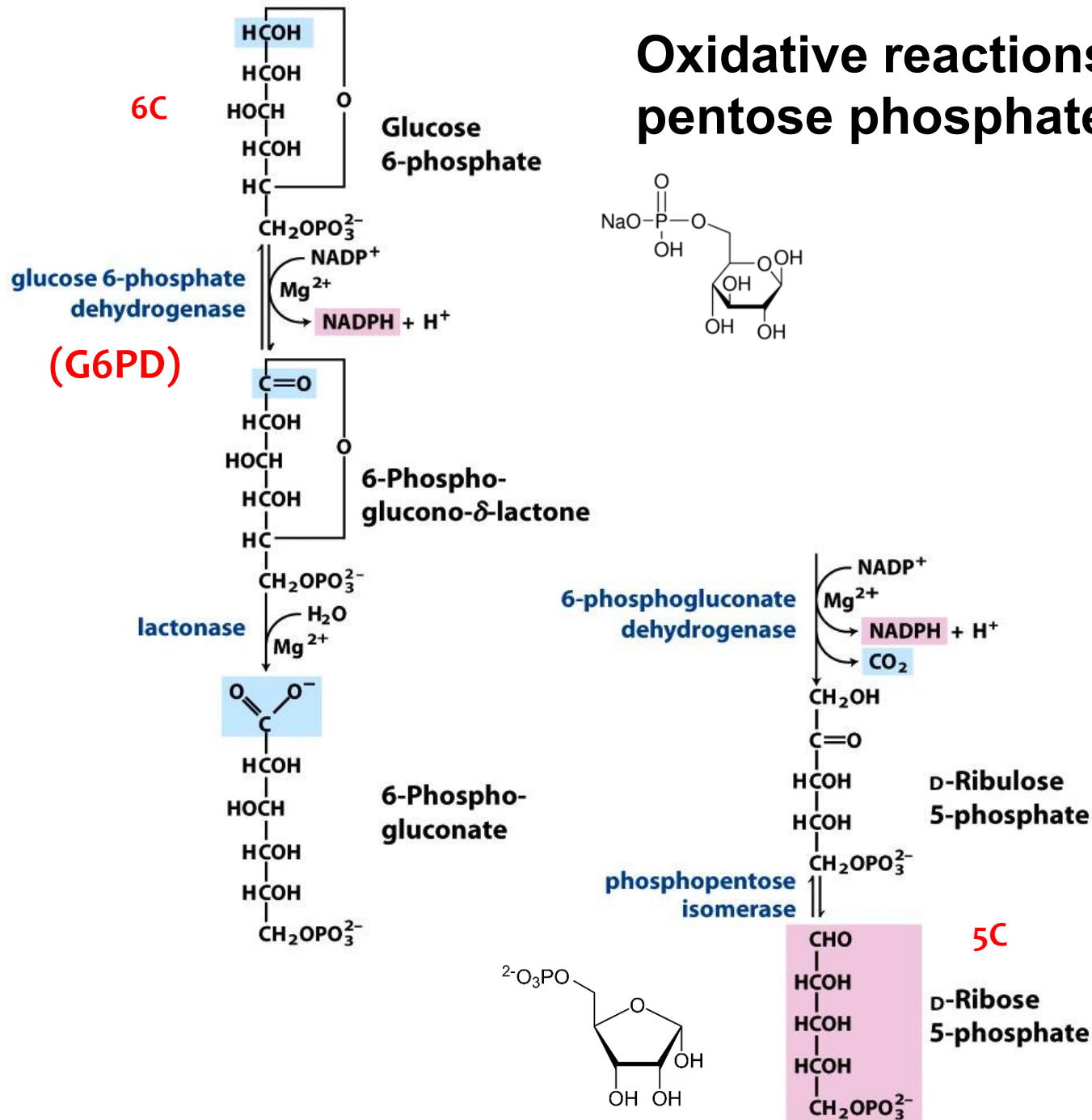
The pentose phosphate pathway (or phosphogluconate pathway)



MESH1 inhibition = Ferroptosis inhibition = Cell survival
GPX4 inhibition = Ferroptosis = Cancer cell death

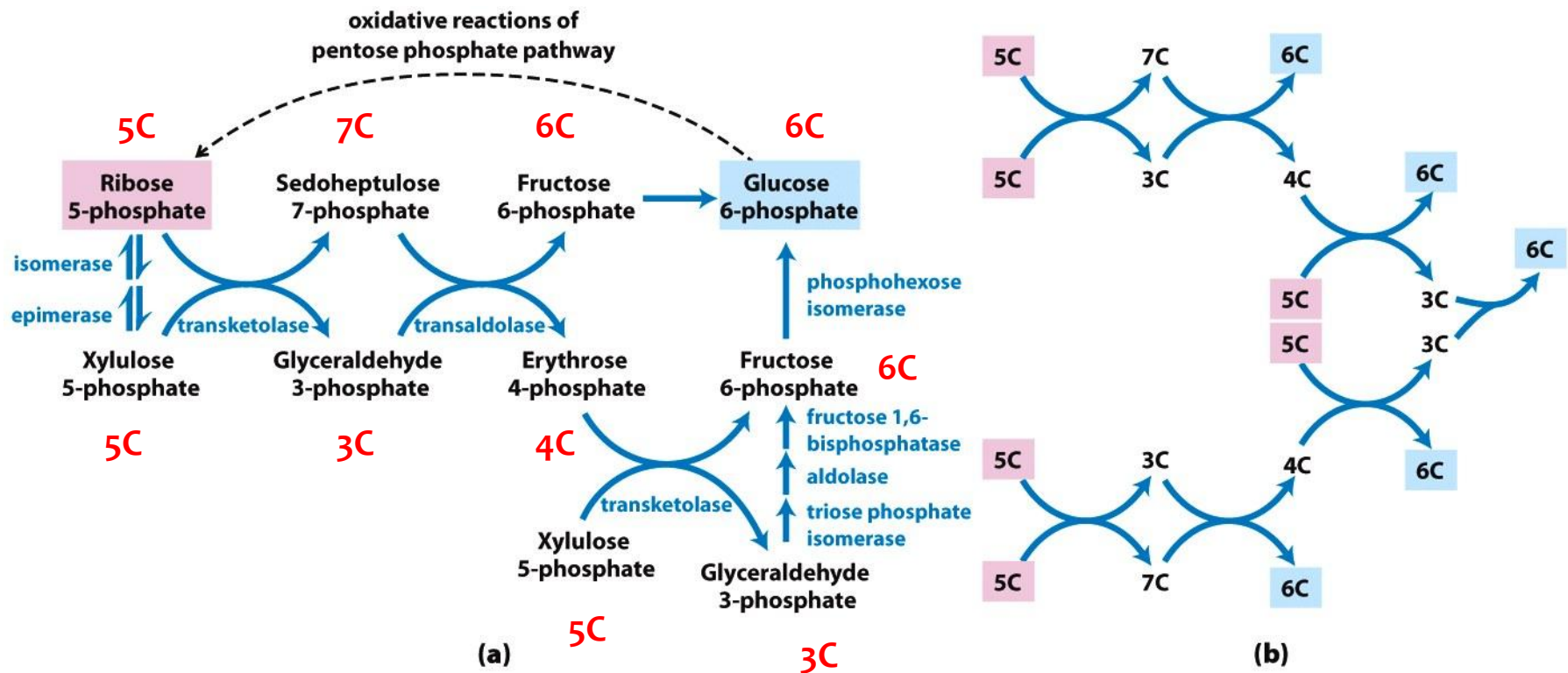


Oxidative reactions of the pentose phosphate pathway



Nonoxidative reactions of the pentose phosphate pathway

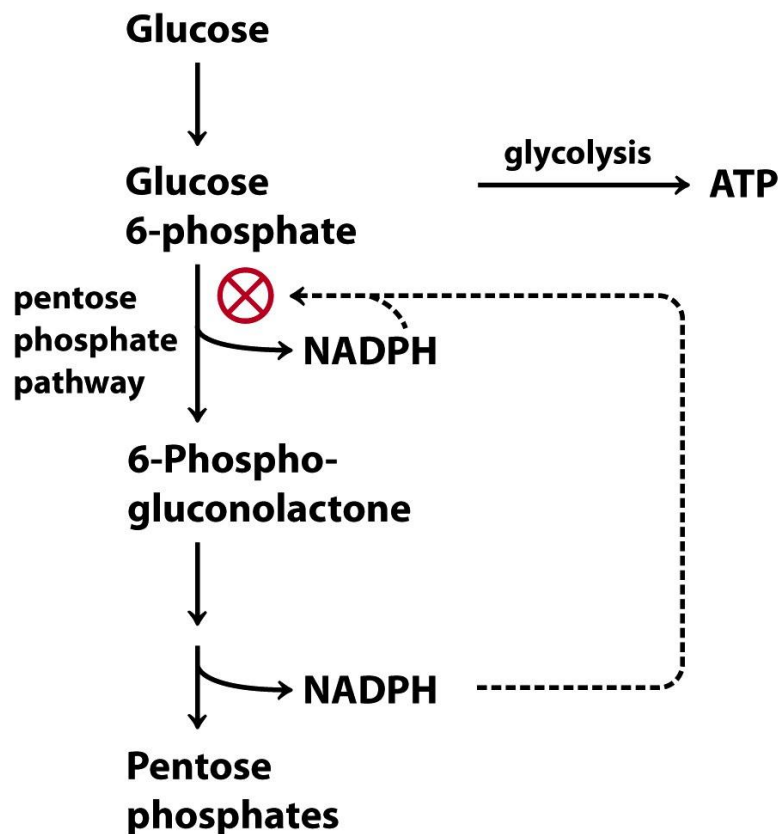
In tissues that require primarily NADPH, the R5P are recycled !



TPP (thiamine pyrophosphate) is needed as a coenzyme !

six 5C → five 6C

Glycolysis or PPP ? It depends on NADPH.

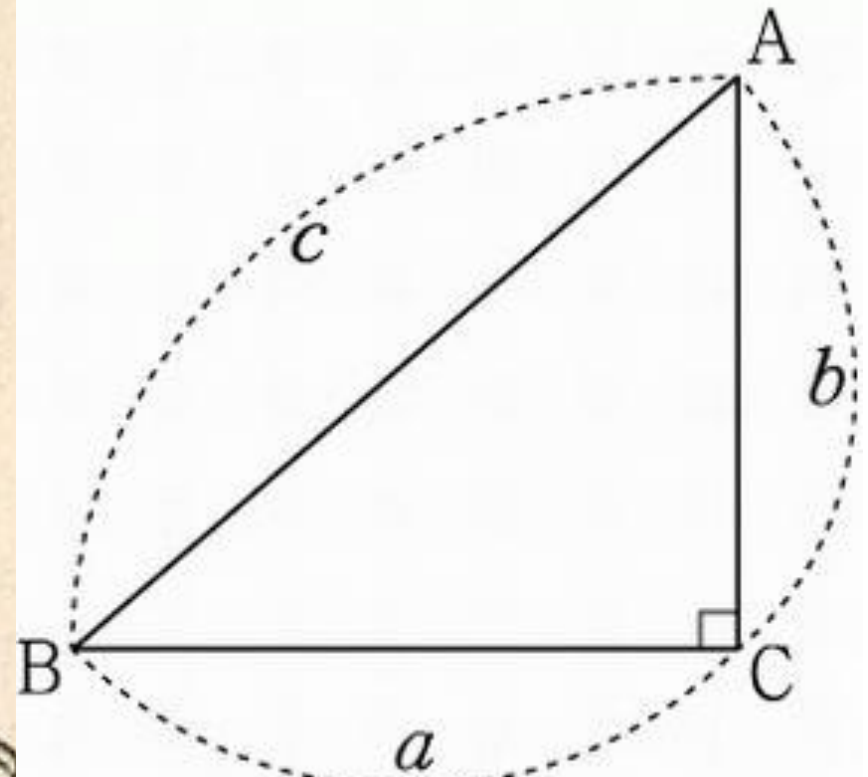
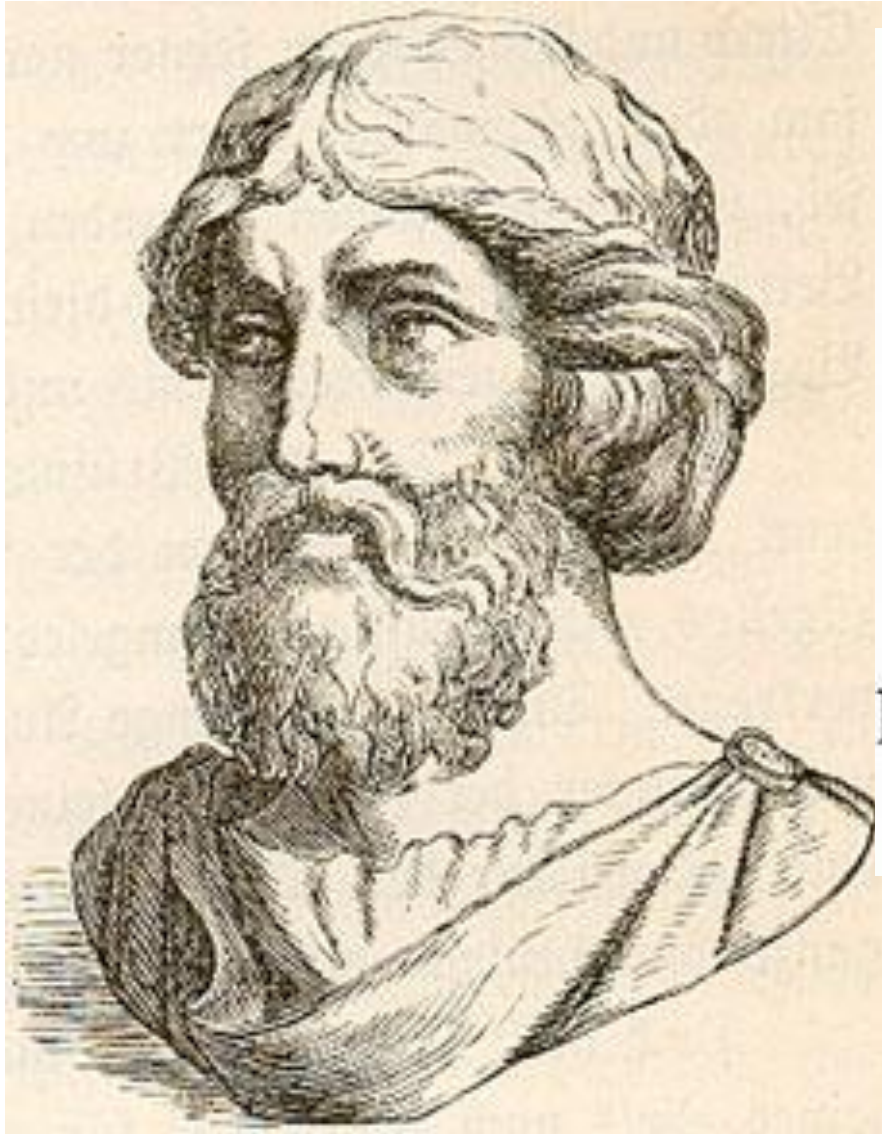


glycolysis, pentose-P pathway,
gluconeogenesis are carried out in cytosol
: shared enzyme & intermediates

NADPH increase $\rightarrow$ glycolysis
NADP⁺ increase $\rightarrow$ pentose-P path

Deficiency in G6PD
(glucose-6-phosphate dehydrogenase)

Pythagoras prohibited his followers from dining on fava beans !



<http://blog.naver.com/sooeasy>

Falafel

FAVA BEANS

Long Pod
Cool Weather Broad Bean

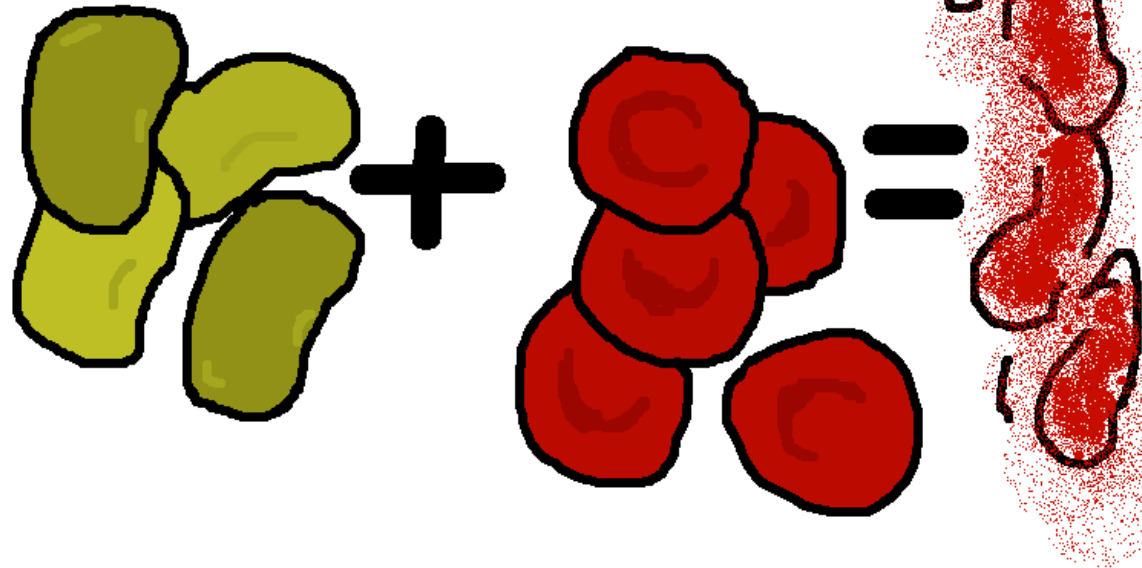
NET WT. 1.5 oz

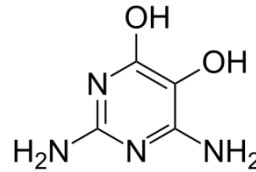


Ed
Hume
Seeds ...



Favism





Divicine from fava beans

→ Divicine reacts with oxygen in red blood cells, which creates ROS (reactive oxygen species)

→ Cause glucose-6-phosphate dehydrogenase (G6PD) deficiency

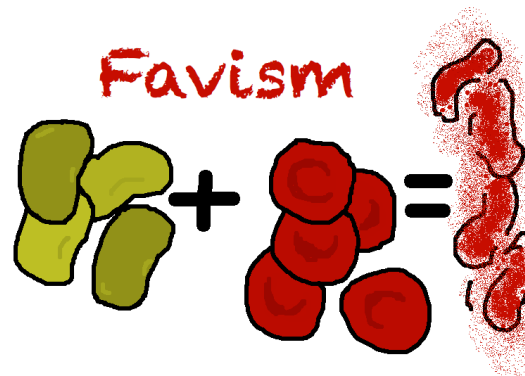
→ pentose-P pathway decreased

→ NADPH production diminished

→ Detoxification of H_2O_2 is inhibited

→ Lipid peroxidation

→ Breakdown of erythrocyte membrane, cellular damage

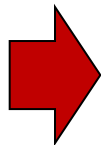


However ...

G6PD deficient human (25% of tropical african)
: resistance to **malaria**

Malaria parasite, *Plasmodium falciparum*, is very
sensitive to **oxidative damage**

Primaquine (antimalarial drug) cause oxidative stress to
the parasite
→ can cause human illness **favism**



To be Favism or Malaria ?
That is the question !!



Wernicke-Korsakoff Syndrome :

Thiamine (**VitB1**) deficiency

Heavy alcohol consumption inhibits thiamine absorption

Mutation in transketolase : low affinity for TPP

- Low level of available TPP
- Slow down of the pentose-P pathway (non-oxidative)
- Wernicke-Korsakoff Syndrome
- Memory loss, mental confusion, and partial paralysis

Summary

Pentose phosphate pathway : $G6P \rightarrow$ pentose-P

(1)oxidative steps: $G6P \rightarrow$ ribulose-5P / $NADP^+ \rightarrow$ NADPH

(2)Nonoxidative steps : pentose-P $\rightarrow$ G6P / 6 pentose $\rightarrow$ 5 hexose

(3)Regulation of pentose-P pathway : by conc of NADPH/ $NADP^+$